

Chapman's Reflexes

John D. Capobianco, D.O., F.A.A.O.

Professor

Department of OMM

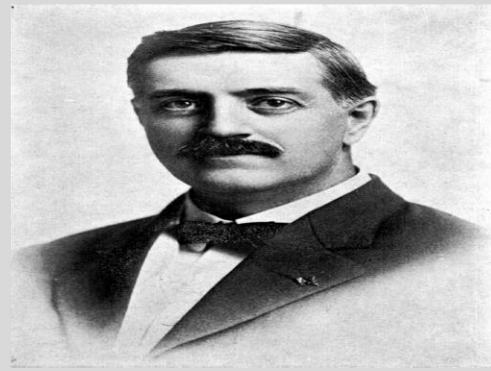


Session Objectives

- Understand that the neurophysiologic mechanism of Chapman's points involves viscerosomatic reflexes
- Relate Chapman's reflexes to the neuroendocrine immune response.
- Describe the common manifestations of endocrine diseases as they relate to Chapman's reflexes.
- Know that Chapman's reflexes involve the sympathetic arm of the autonomic nervous system.
- Use memory tools for Chapman's points such as high density areas like the second intercostal space and the lateral thigh.
- Incorporate Chapman's reflexes into the five models of osteopathic medicine.



Frank Chapman, D.O.



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

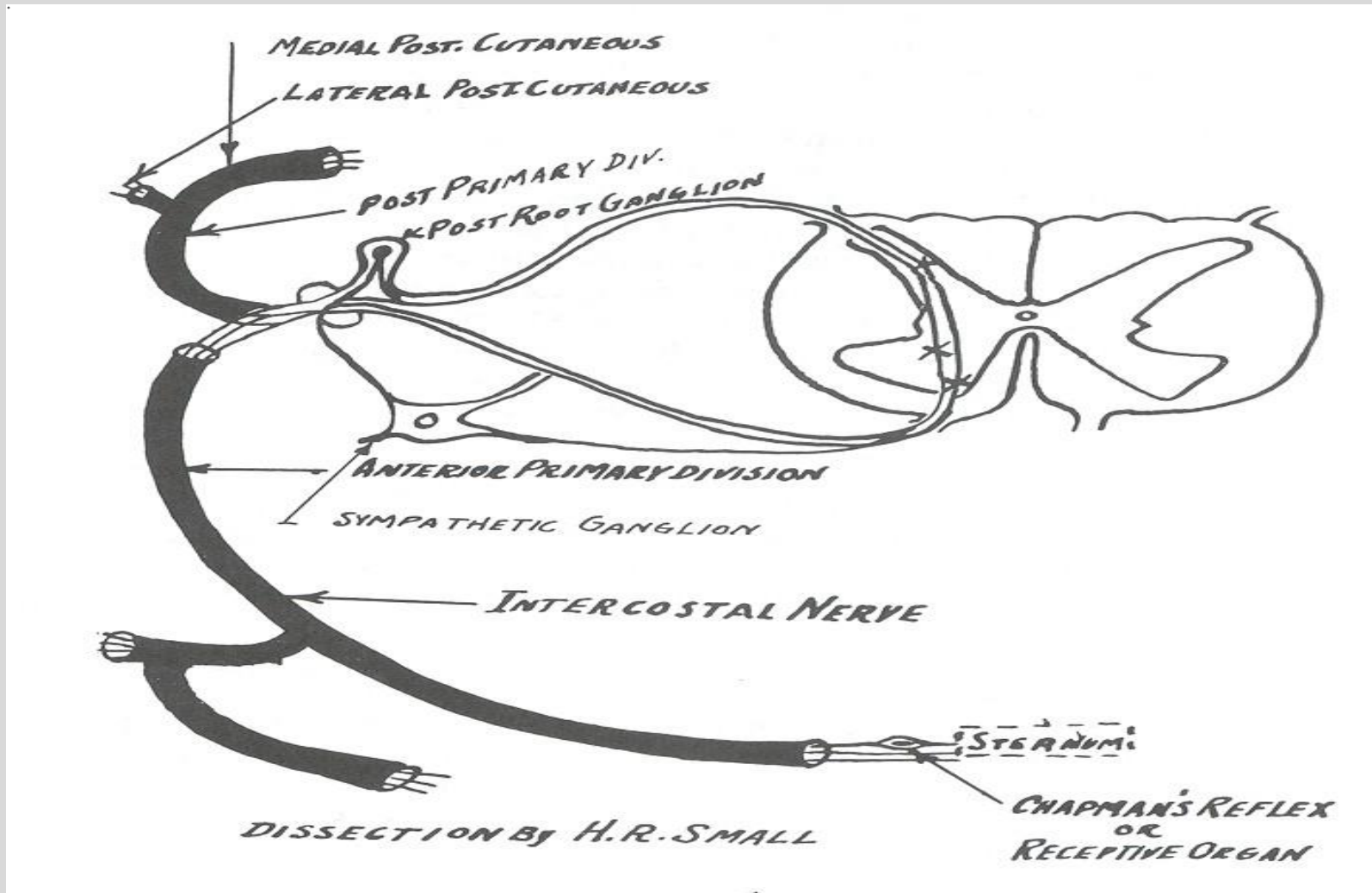
- At the core of classic osteopathy as taught by his teacher, A.T. Still, was the concept of diagnosing autonomic tone by the “feel of the tissues” and recognizing the unlimited potential of the lymphatics.
- Dr. Frank Chapman practiced and charted his “lymphatic reflexes” for over thirty years.
- Upon Dr. Frank Chapman’s death, his wife, Dr. Ada Hinckley Chapman, and his brother-in-law, Dr. Charles Owens along with Dr. W.F. Link assembled his writings into “An Endocrine Interpretation of Chapman’s Reflexes”. This set the stage for the concept of unity of hormones, immunity and the autonomic nervous system.



Chapman's Reflex Receptive Organ

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

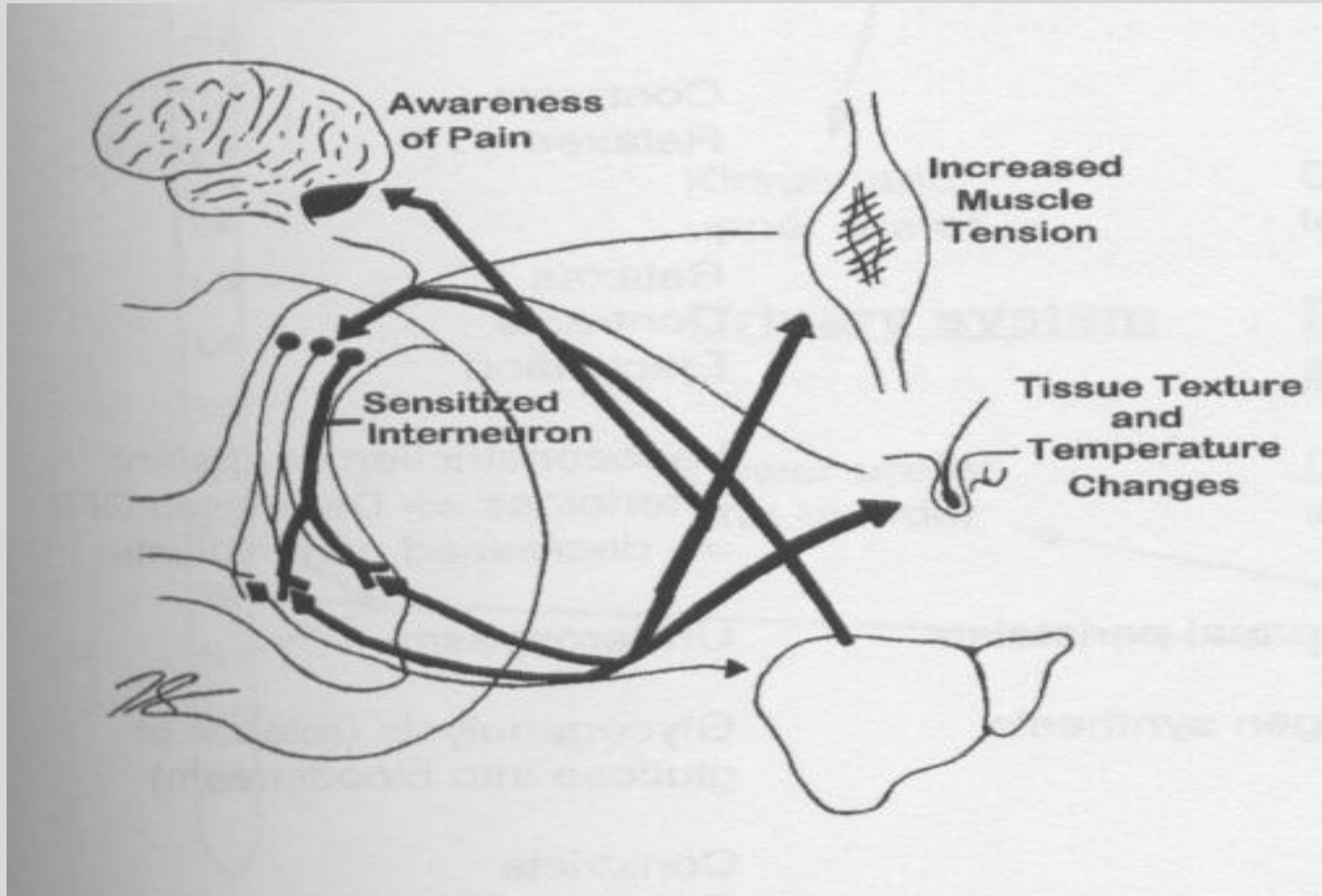


Viscerosomatic Reflex

Chapman's Reflexes are Viscerosomatic Reflexes

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Chapman's Reflex Points: “Gangliform” Contractions

Hypercongestion from hypersympathetic tone to deep intercostal fascia resulting in lymphatic edema and myofascial thickening. This is reflected as a painful nodule that is described by Patriquin as:

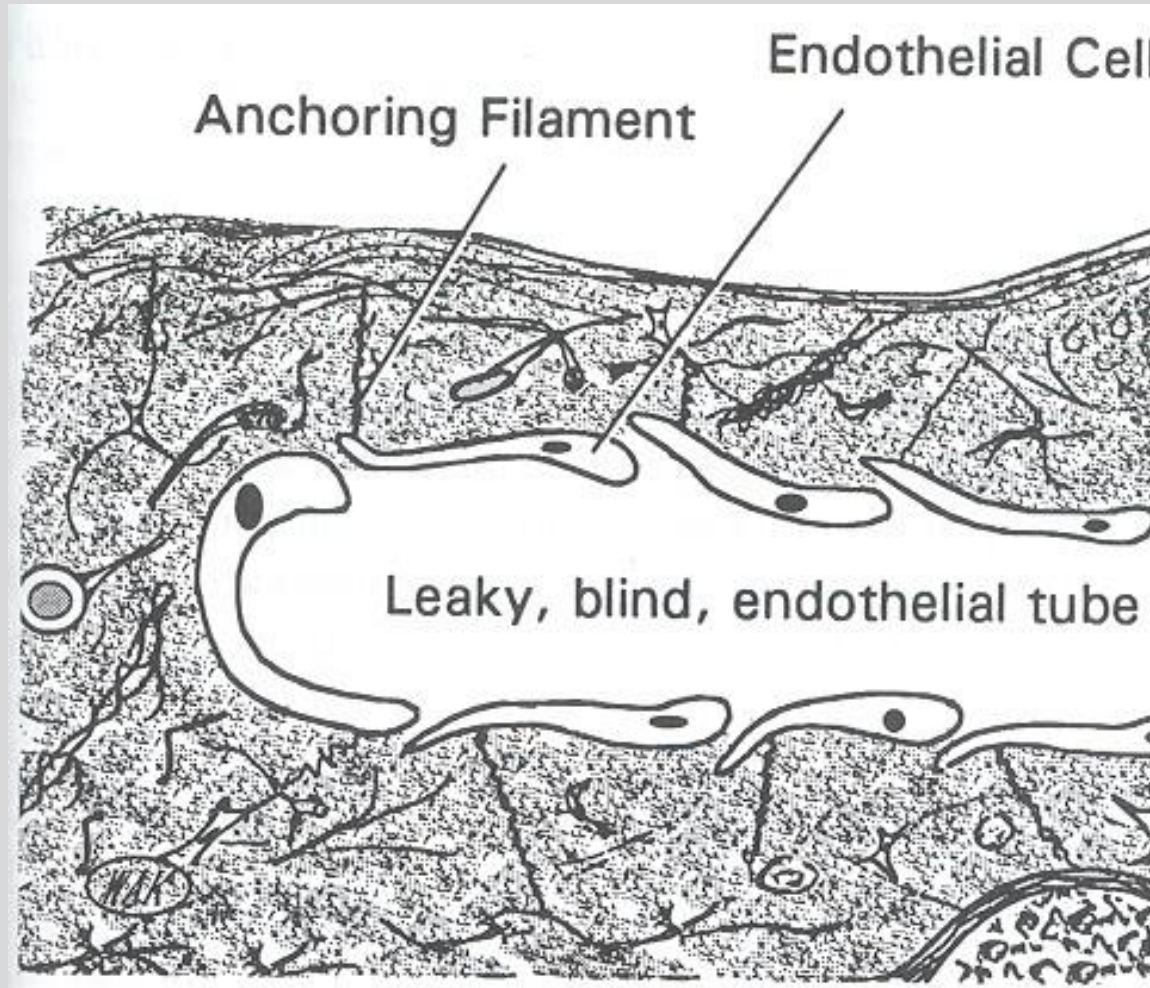
- Small
- Smooth
- Firm
- Discreetly palpable
- Approximately 2-3mm in diameter

Note: Posterior points may feel like any other SD

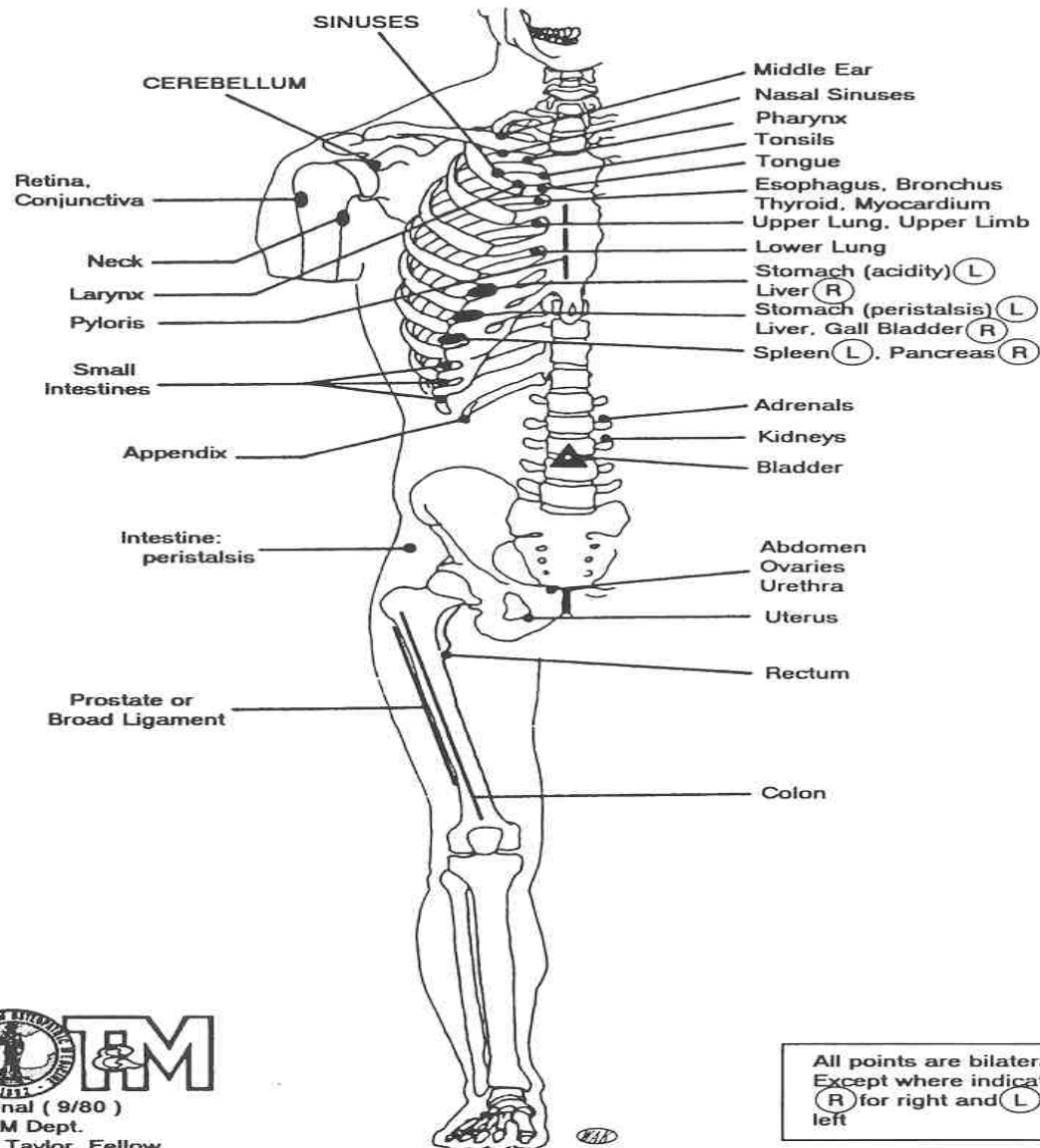
Gangliform contraction that blocks lymphatic drainage causing inflammation in tissues distal to the blockage.

- Lies deep to the skin lying on the deep fascia or periosteum
- Typically paired reflexes on the front and back side of the body
- Fixed at specific anatomic locations

Beginning of Lymphatic System



CHAPMAN'S REFLEXES: anterior points



Original (9/80)
OT&M Dept.
Reid Taylor, Fellow

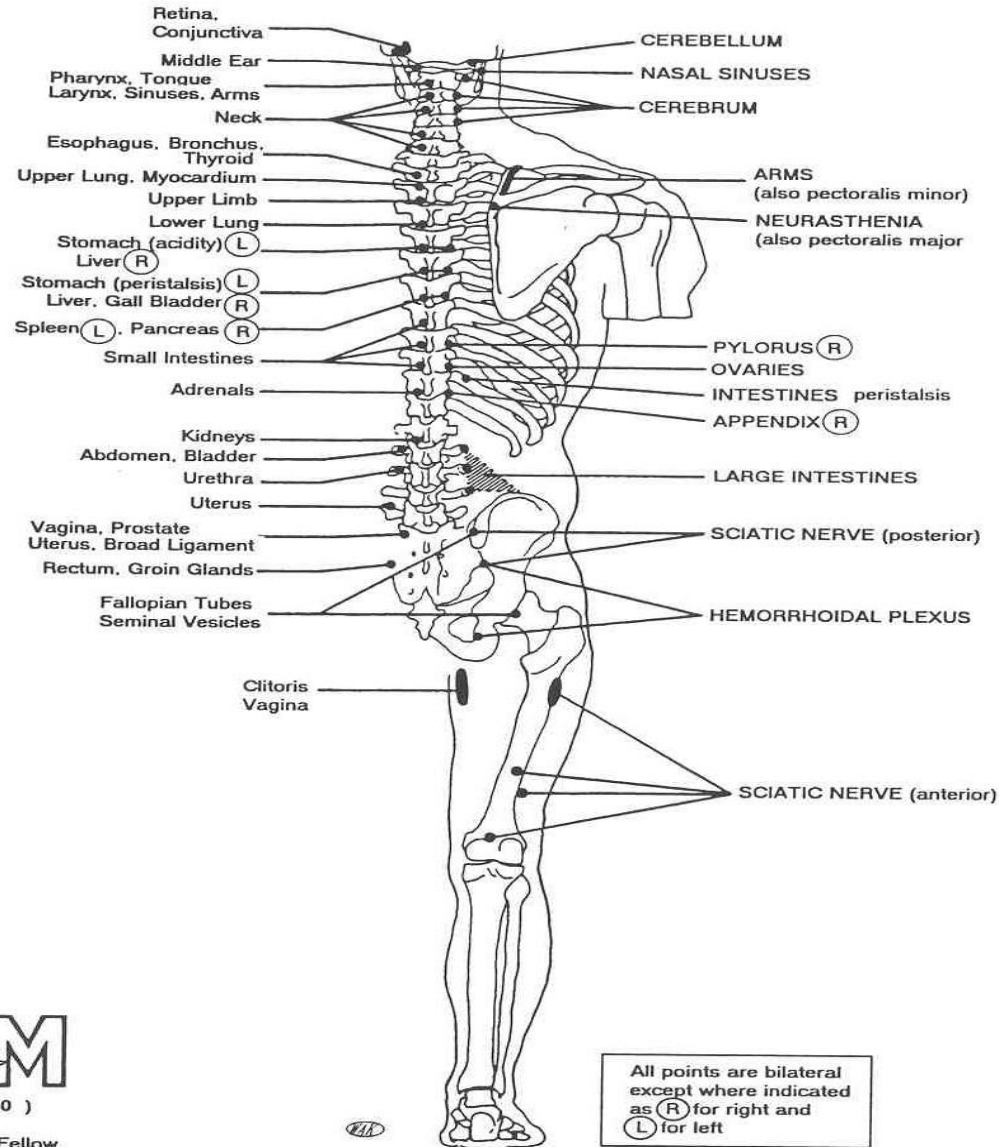
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Anterior Chapman's Reflexes



CHAPMAN'S REFLEXES: posterior points



Original (9/80)
OT&M Dept.
Reid Taylor, Fellow

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

- Located between spinous processes & the transverse processes below
- Not as well circumscribed as anterior reflexes
- Feels more like a somatic dysfunction



High Density Areas of Chapman's Reflexes

2nd Intercostal Space (ICS):

- Bronchus (B)
- Esophagus (E)
- Thyroid (T)
- Heart (H)
- B.E.T.H.
- Clinical correlations include a patient with bronchitis who is status post MI with a history of GERD and Hashimoto's thyroiditis



High Density Areas of Chapman's Reflexes, II

Lateral Femur (Lateral thigh)

- Broad Ligament-Outer Femur (B)
- Prostate-Outer Femur (P)
- Colon-Anterior Outer Femur (C)
- “Sciatic Neuritis” (aka Lumbosacral radiculopathy)-Posterior Outer Femur (S)

“Back Pain Causes Spasm”



- The treatment of these reflexes may alter sympathetic influences on any related organ and therefore disease. As a result, affect the viscerosomatic reflex.
- Diagnostic importance as a result of predictable and consistent reflexes.



Treatment

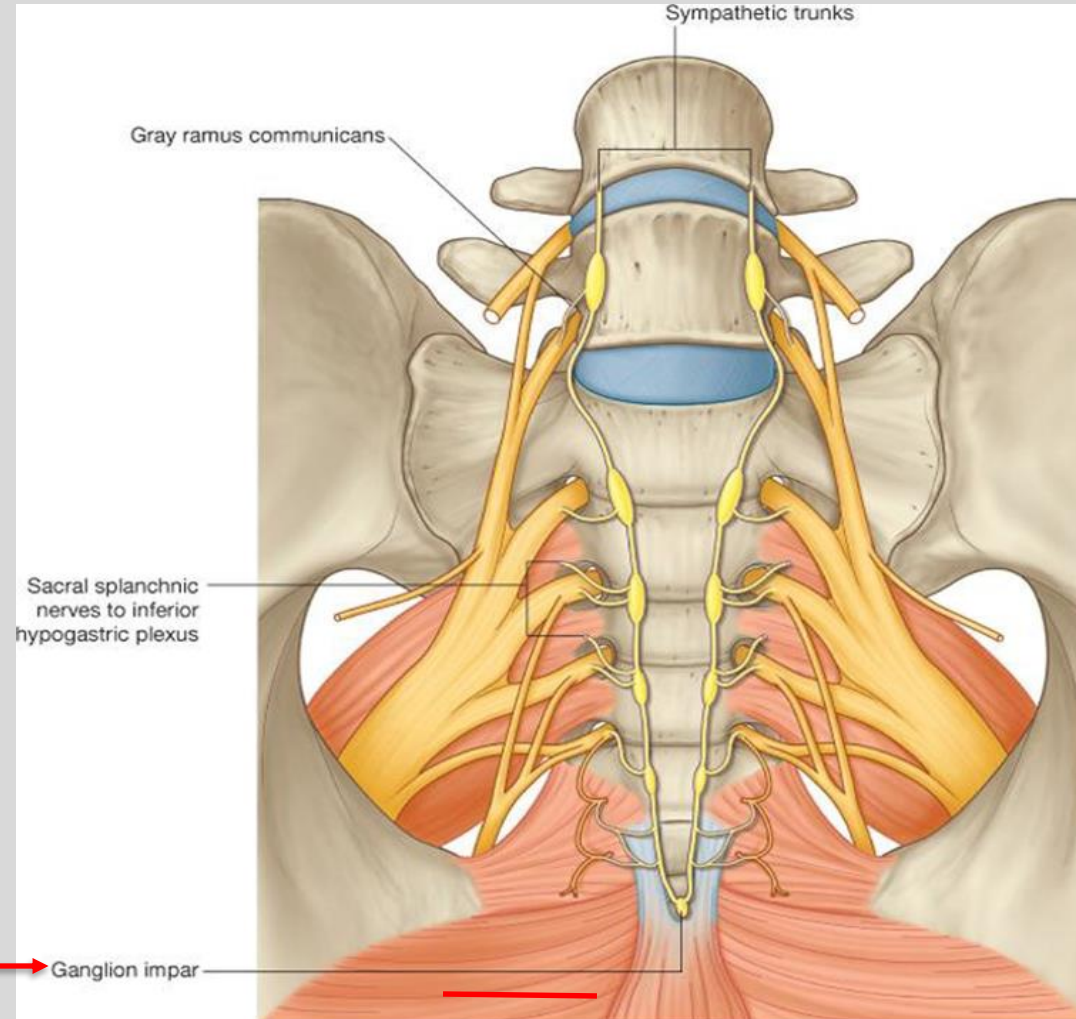
- Pelvis treated first (if systemic problem)
- Next treat anterior points with one finger pad firmly applied directly on gangliform
- Then a rotatory force is applied anywhere from 10 seconds - two minutes
- Treatment completed when nodule resolves and/or marked decrease in pain
- Treat posterior reflexes if necessary



Where does the sympathetic nervous system really end?

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



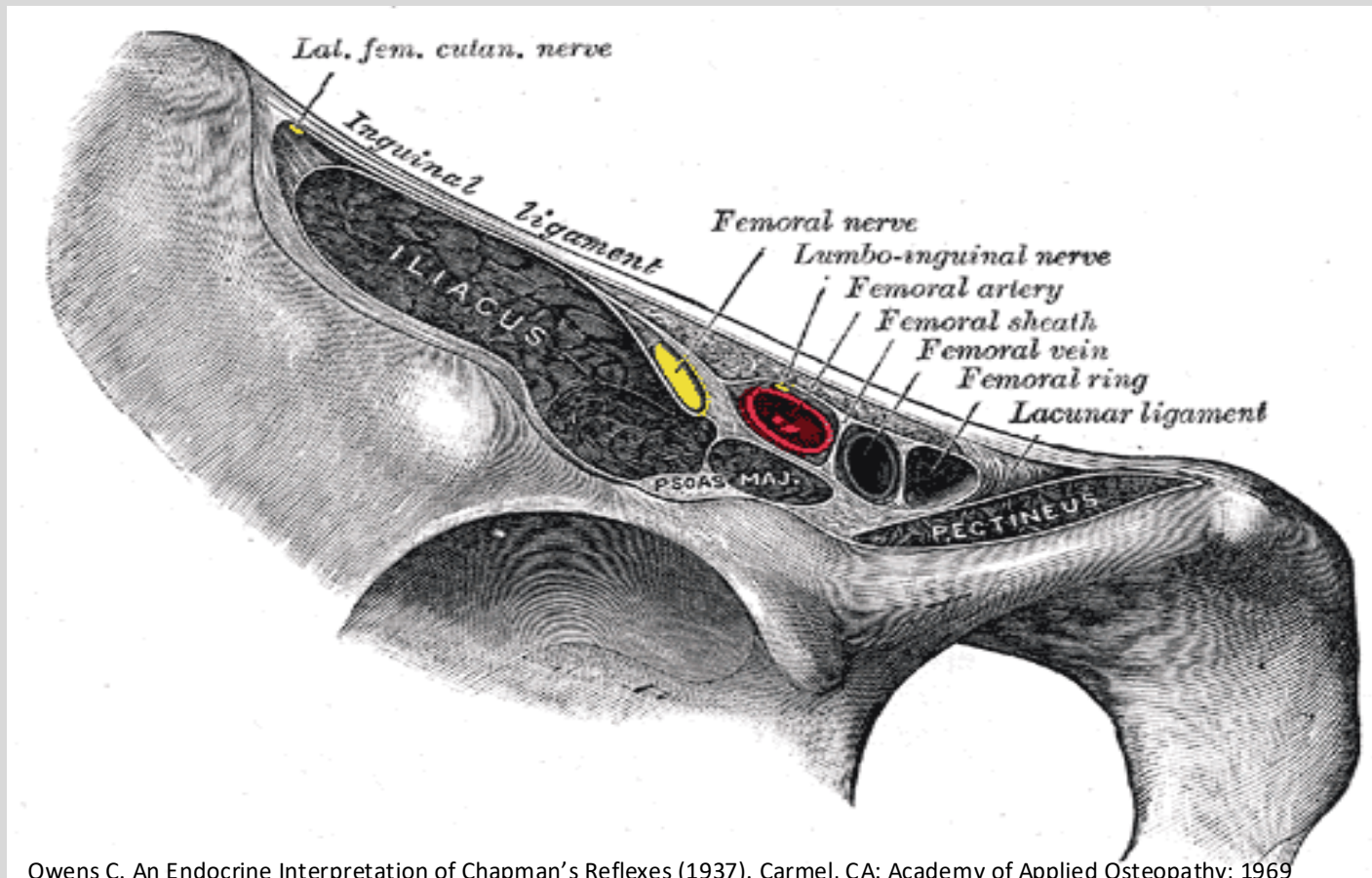
Ganglion Impar is
anterior to the sacrococcygeal joint

Drake: Gray's Anatomy for Students, 2nd Edition.
Copyright © 2000 by Churchill Livingstone, an imprint of Elsevier Inc. All rights reserved.

[Chapter 22: DiGiovanna](#)



Look for tensility of the inguinal ligament to
look for innominate/pelvic dysfunction



Owens C. An Endocrine Interpretation of Chapman's Reflexes (1937). Carmel, CA: Academy of Applied Osteopathy; 1969



From: Muscle Energy

An Osteopathic Approach to Diagnosis and Treatment, 3e, 2004



Legend: The Muscle Energy model of innominate correction, especially position, used even in a passive fashion, would suffice to correct a pelvic dysfunction. The heritage of Dr. Fred Mitchell, Sr and Jr. to the Chapman's model goes beyond a power point presentation



Research

- Treatment of posterior points (T10-11):
- Decreased systolic/diastolic blood pressure by 15mmHg/8mmHg
- Decreased serum aldosterone

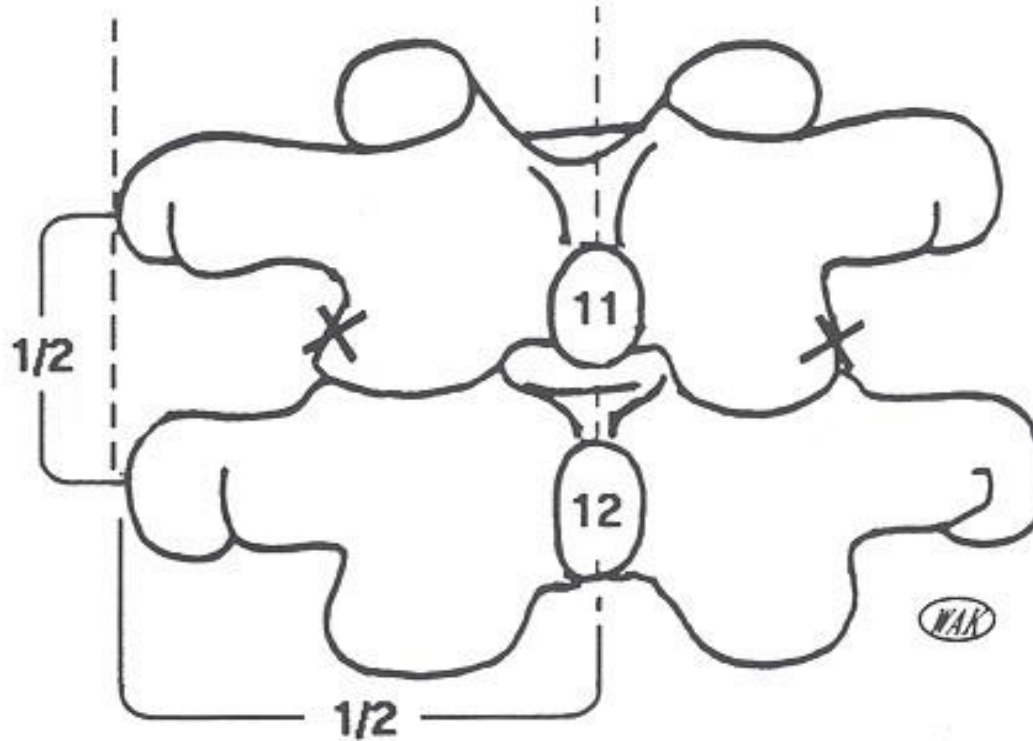


Hypertensive Patients

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Posterior Adrenal
Chapman's Reflex



Rotatory stimulation of the posterior Chapman's points (T11-12) depicted above have been shown to effectively lower blood pressure and decrease serum aldosterone levels.³⁸



Chapman's Reflexes and the hospitalized patient with pneumonia:

- Statistical for the presence of Chapman's Reflexes in the third & fourth intercostal space and between T3-4 and T4-5



Case Study

Nephrotic Syndrome

- 19 yr. Old AA female with nephropathy who presented with oliguria and peripheral edema.
- Unresponsive to diuretics, albumin and dietary restrictions.
- Daily urine output about 250 cc's
- OMT given at day 5 of hospitalization.
- OMT directed to lymph, diaphragms and Chapman's point at the periumbilical and thoracolumbar regions
- Immediate 400 cc's diuresis within 45 minutes.
- Next three days patient voided about 1500-2700 cc's of urine.
- Avoided cyclophosphamide.
- Patient was discharged and treated as an outpatient on day 8.

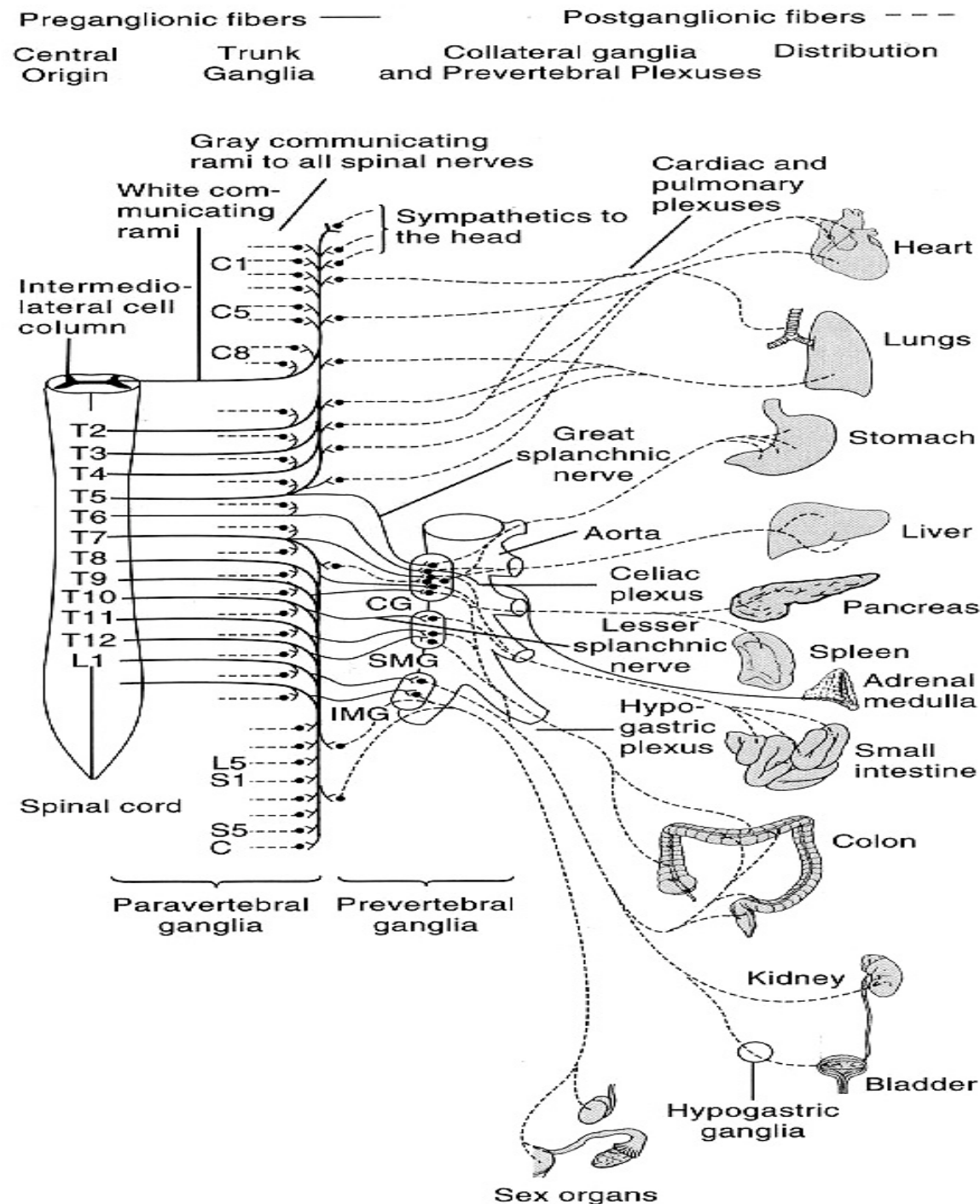


Sympathetic Nervous System

In every disease process there is hypersympathetic activity; sustained sympathetic activity is a common factor in disease.

-IM Korr





NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Sympathetic Innervation from Head through
Pelvis



Upper Respiratory

Bilateral Anterior Points

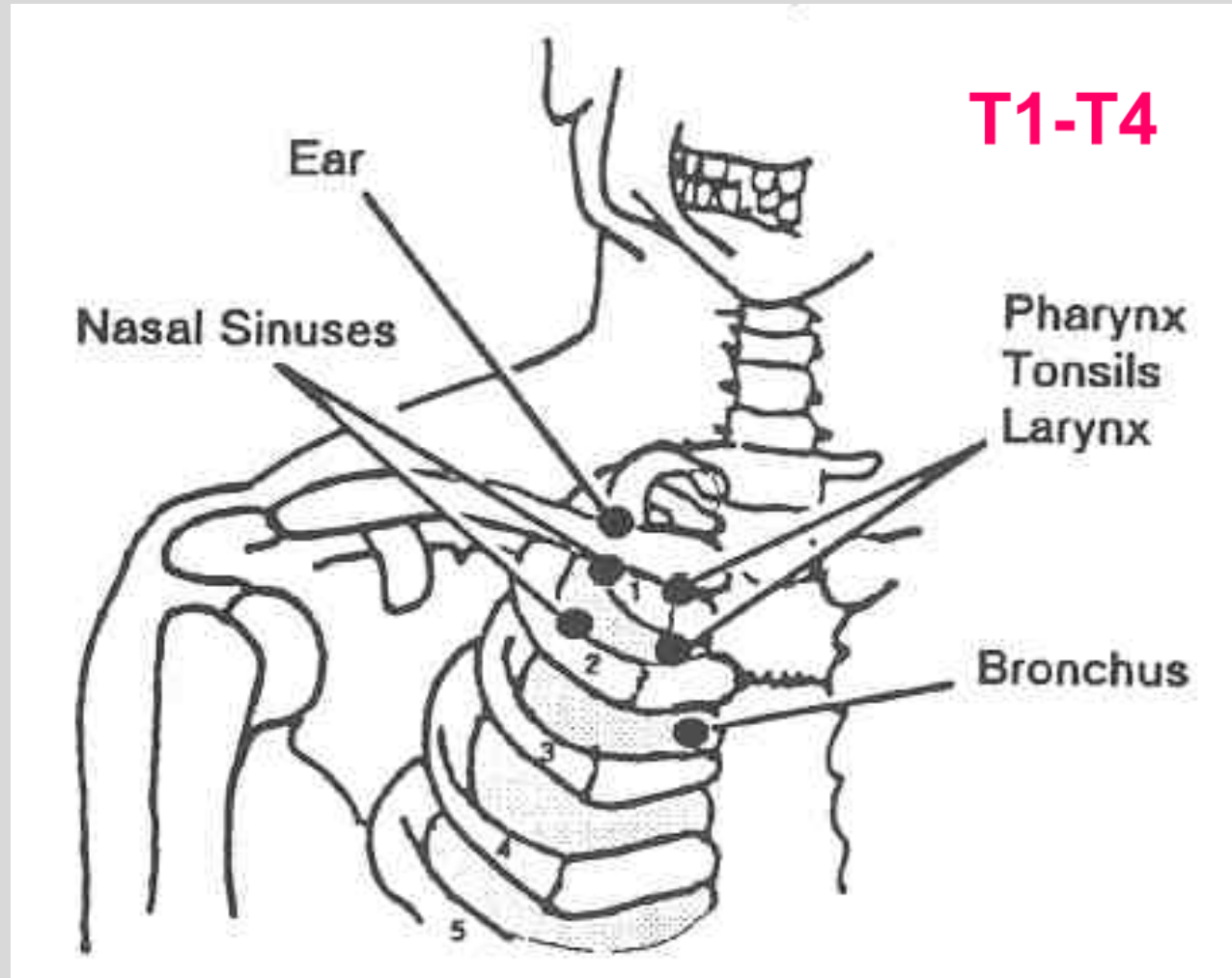
Pharynx: 1st Rib

Tonsils: 1st ICS

Larynx: 2nd Rib

Nasal Sinus: 1st ICS & 2nd Rib

Ear: Upper Edge Medial Clavicle



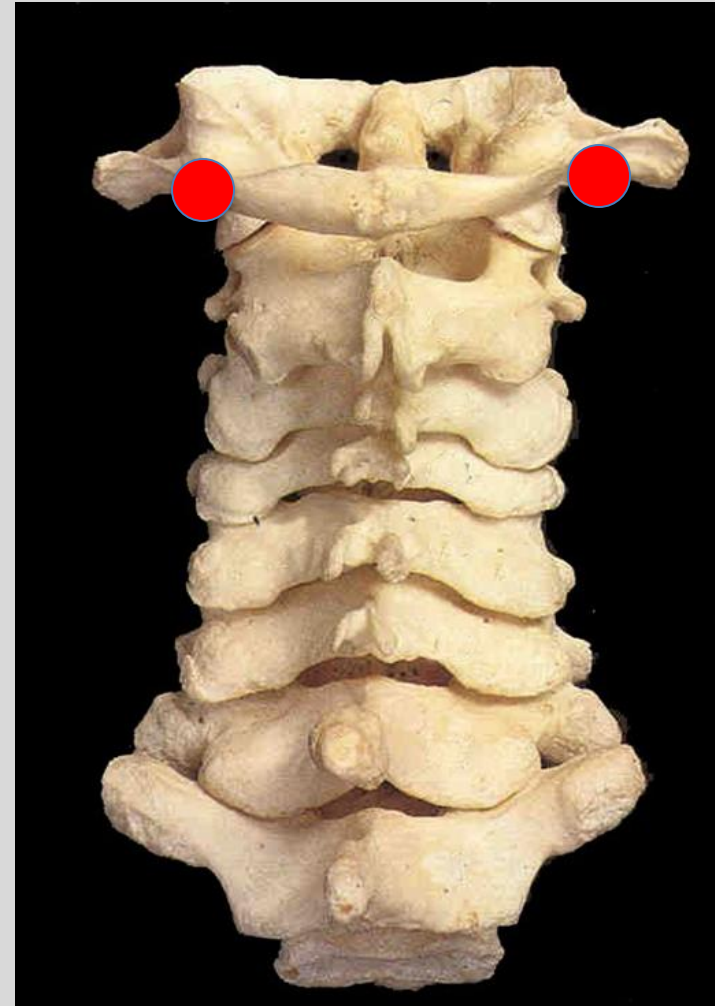
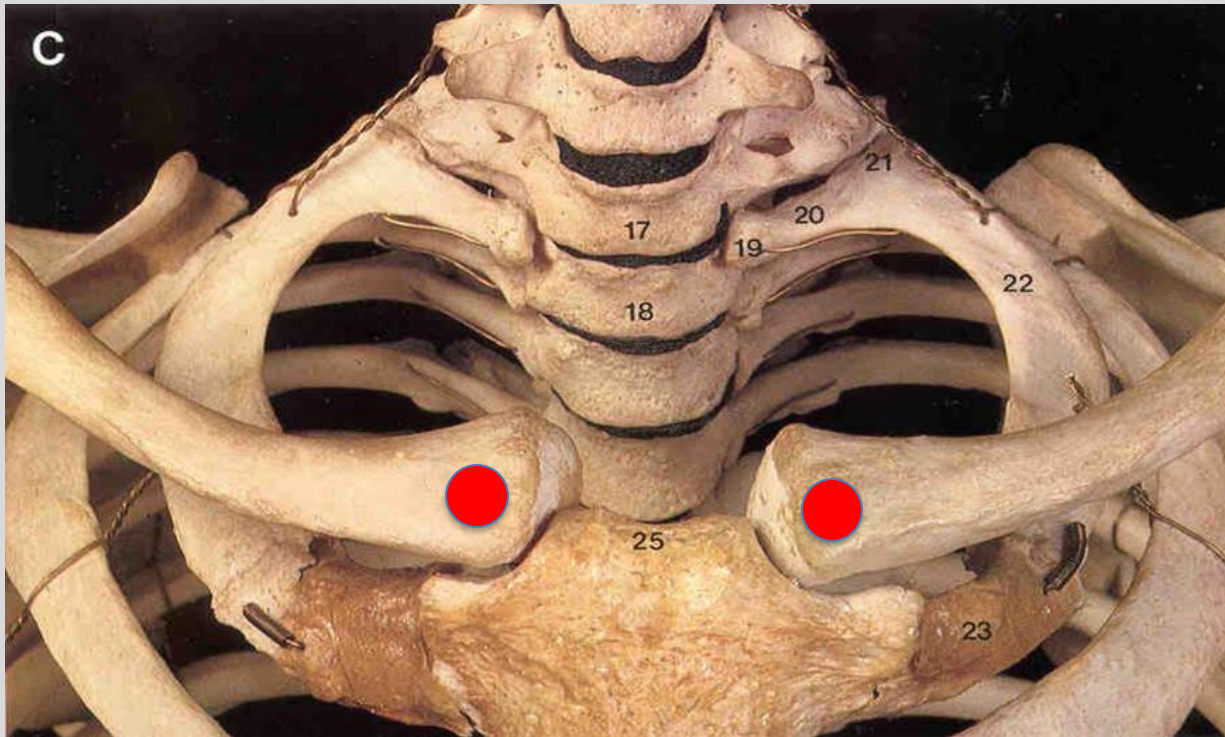
Otitis Media

Anterior: Clavicle upper edge beyond crossing with 1st rib

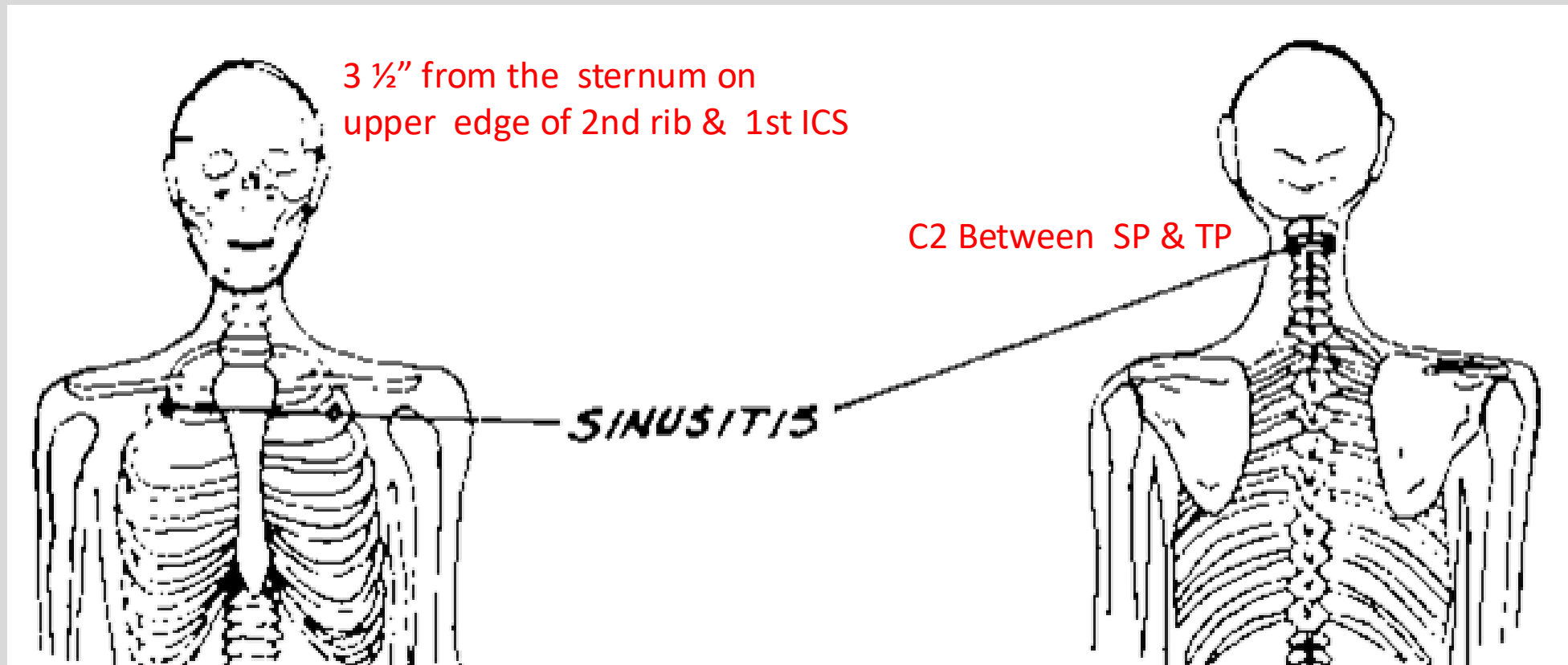
Posterior: Transverse process C1

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



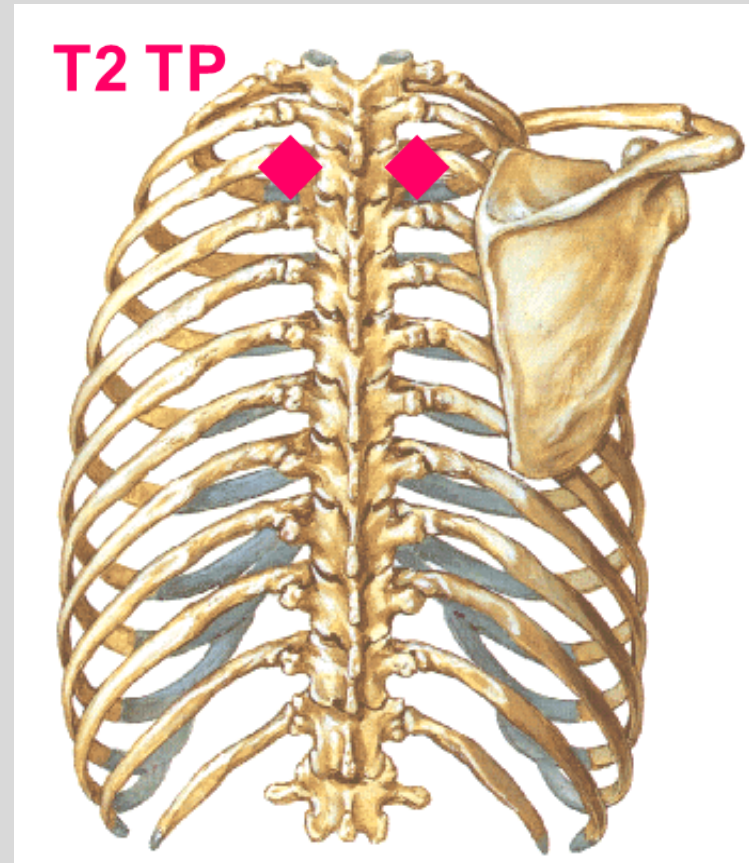
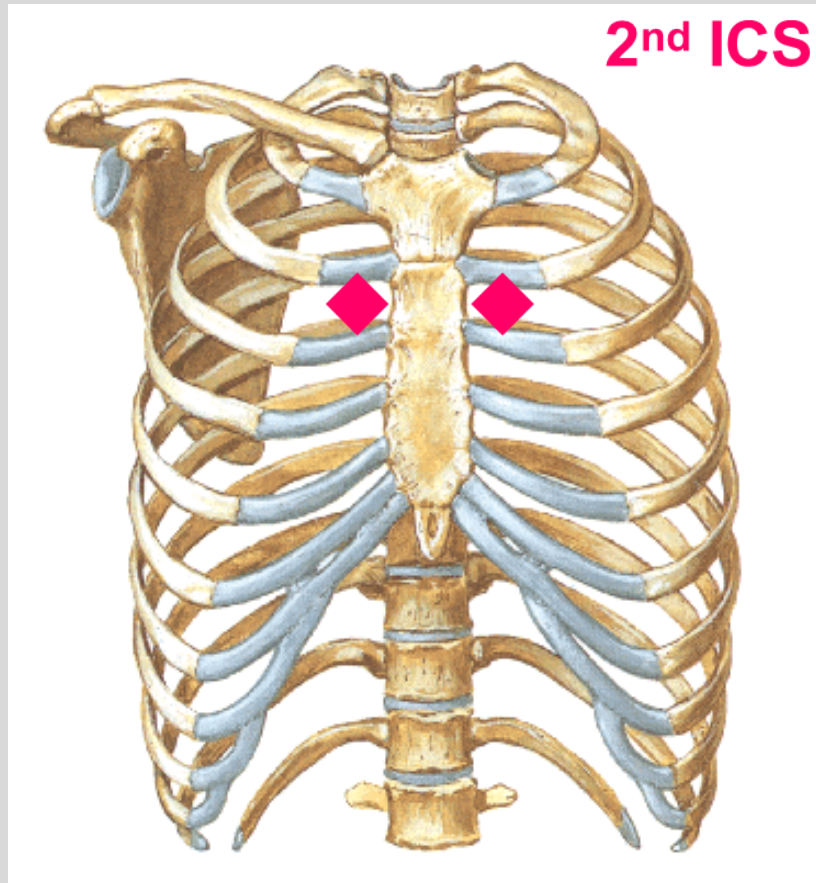
Sinusitis



Thyroid

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Cardiopulmonary

NEW YORK INSTITUTE OF TECHNOLOGY

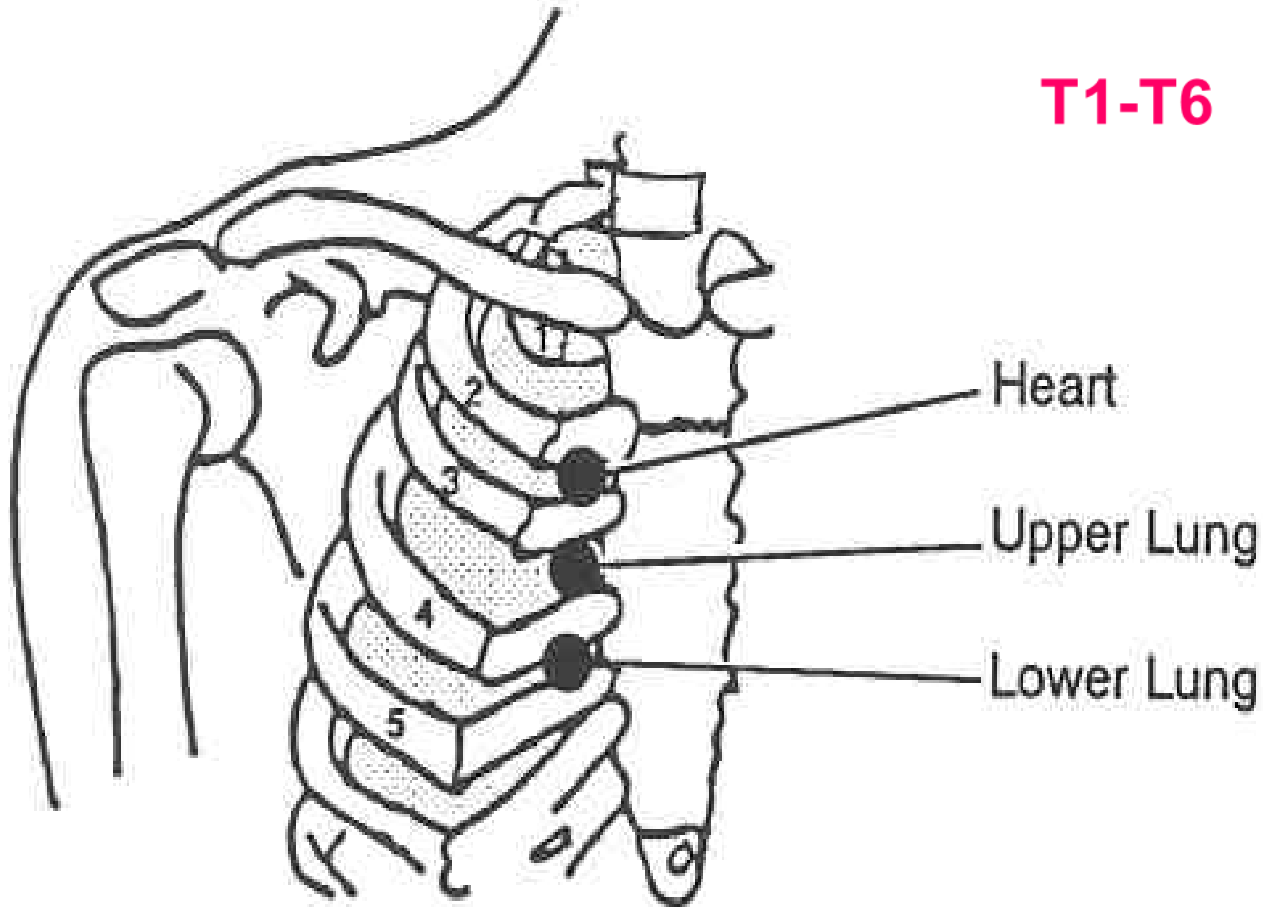
College of Osteopathic
Medicine

Bilateral Anterior Points

Heart: 2nd ICS

Upper Lung: 3^d ICS

Lower Lung: 4th ICS



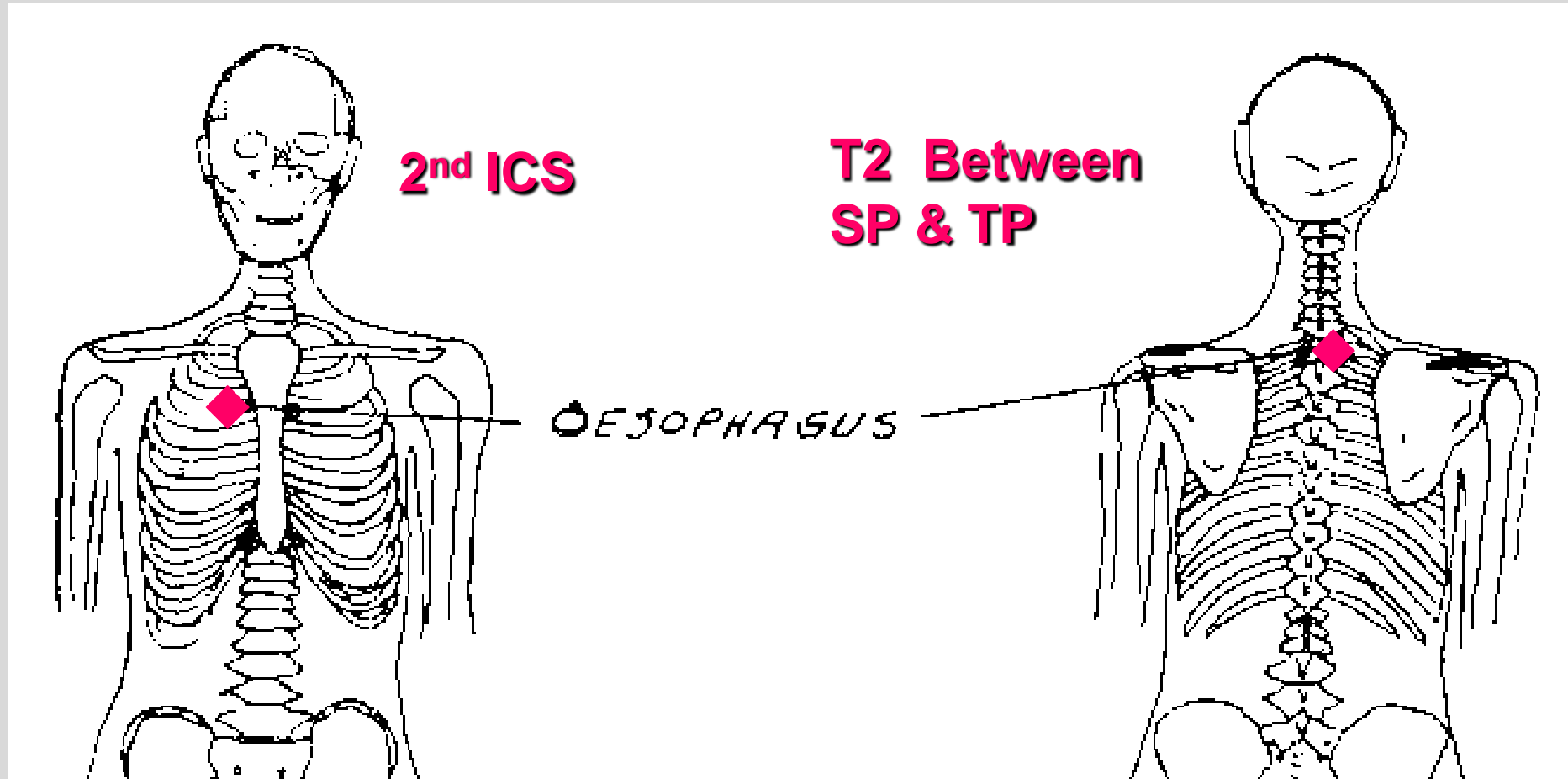
Owens C. An Endocrine Interpretation of Chapman's Reflexes (1937). Carmel, CA: Academy of Applied Osteopathy; 1969



Esophagus (GERD)

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Upper Gastrointestinal

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Anterior Points

Liver: Right 5th ICS and Right 6th ICS

Gallbladder: Right 6th ICS

Pancreas: Right 7th ICS

Stomach (acidity): Left 5th ICS

Stomach (peristalsis): Left 6th ICS

Posterior Points

At same level between the transverse processes
and spinal processes

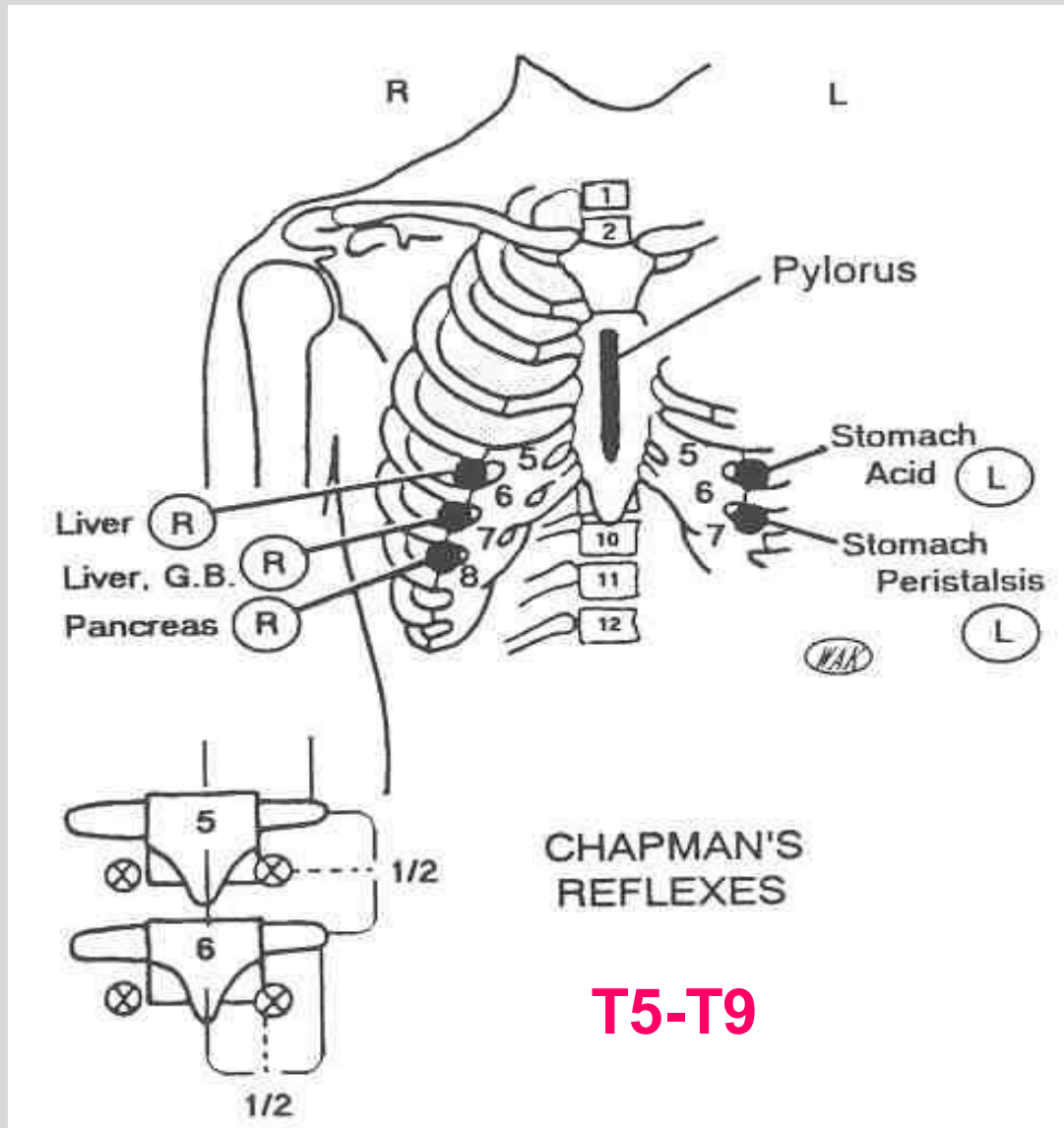
Liver: Right between 5th - 6th and 6th - 7th vertebra

Gallbladder: Right between 6th - 7th vertebra

Pancreas: Left between 7th - 8th vertebra

Stomach: Left between 5th - 6th and 6th - 7th

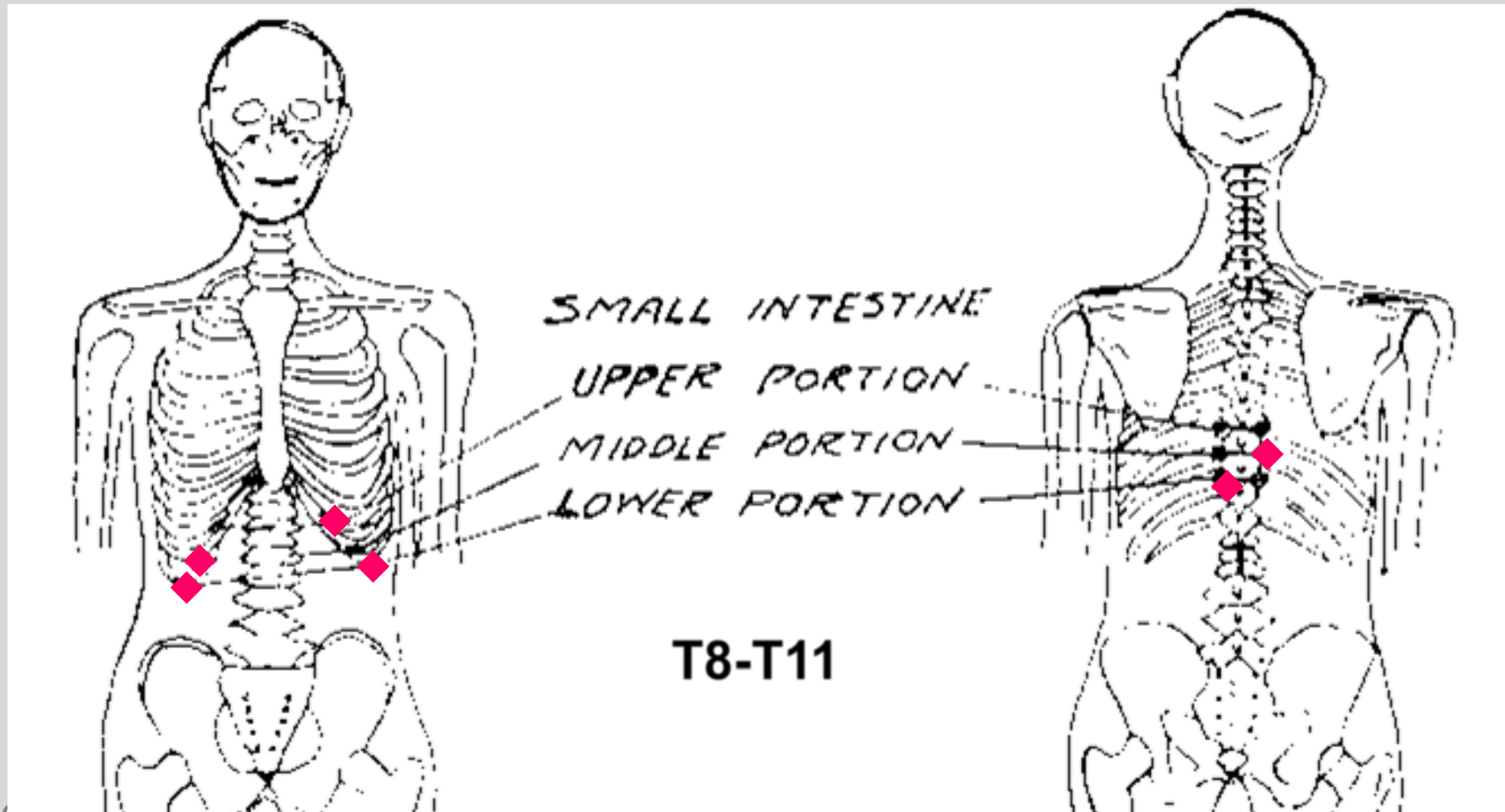
vertebra (acidity and peristalsis respectively)



Small Intestines

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Anterior
8,9,10 ICS

Posterior
Intertransverse
space 8-9, 9-
10, 10-11

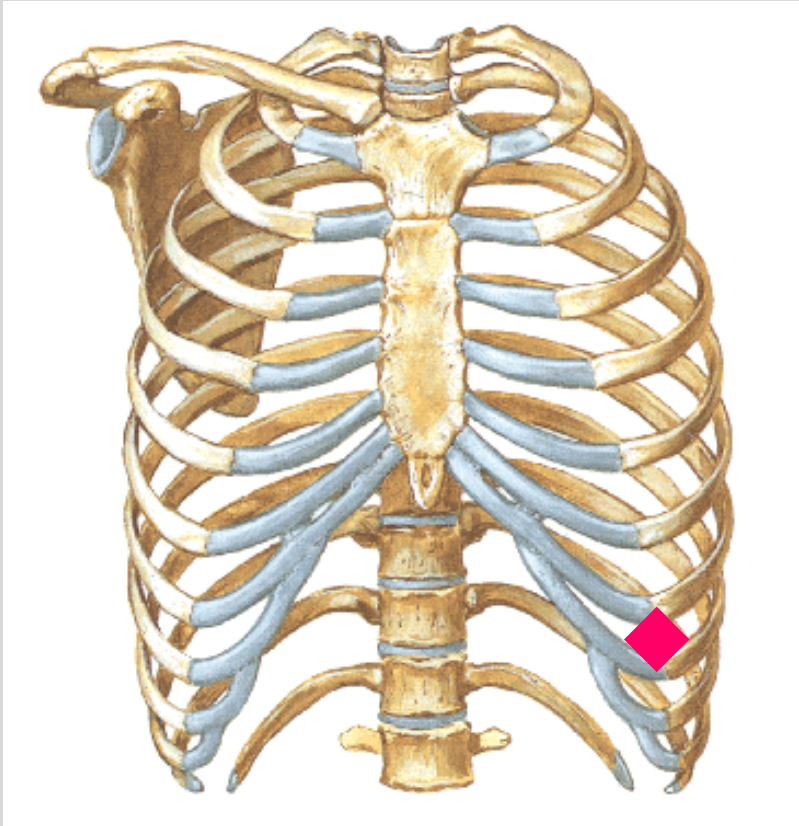


Spleen

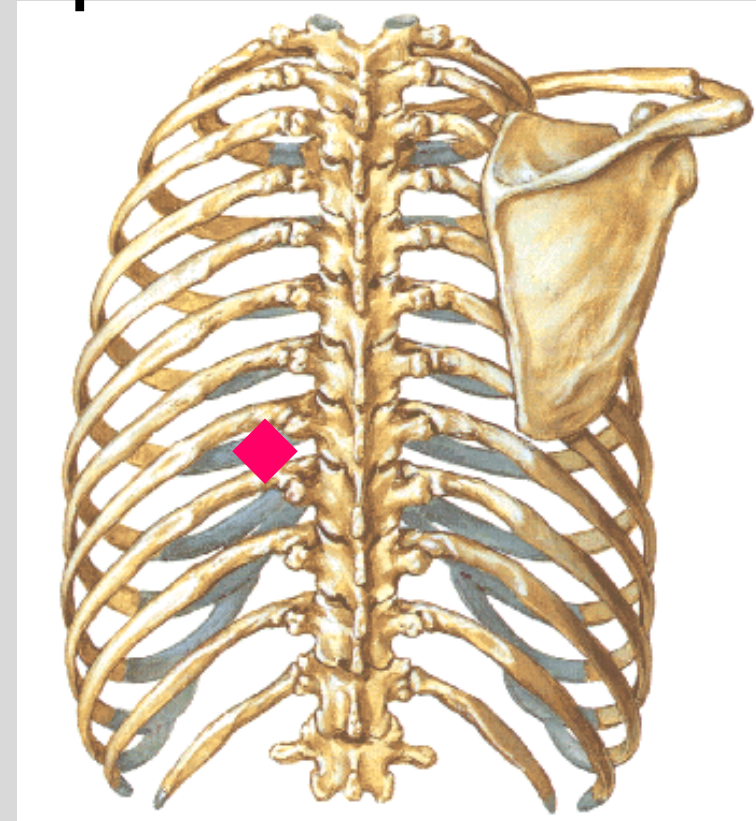
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Left 7th ICS



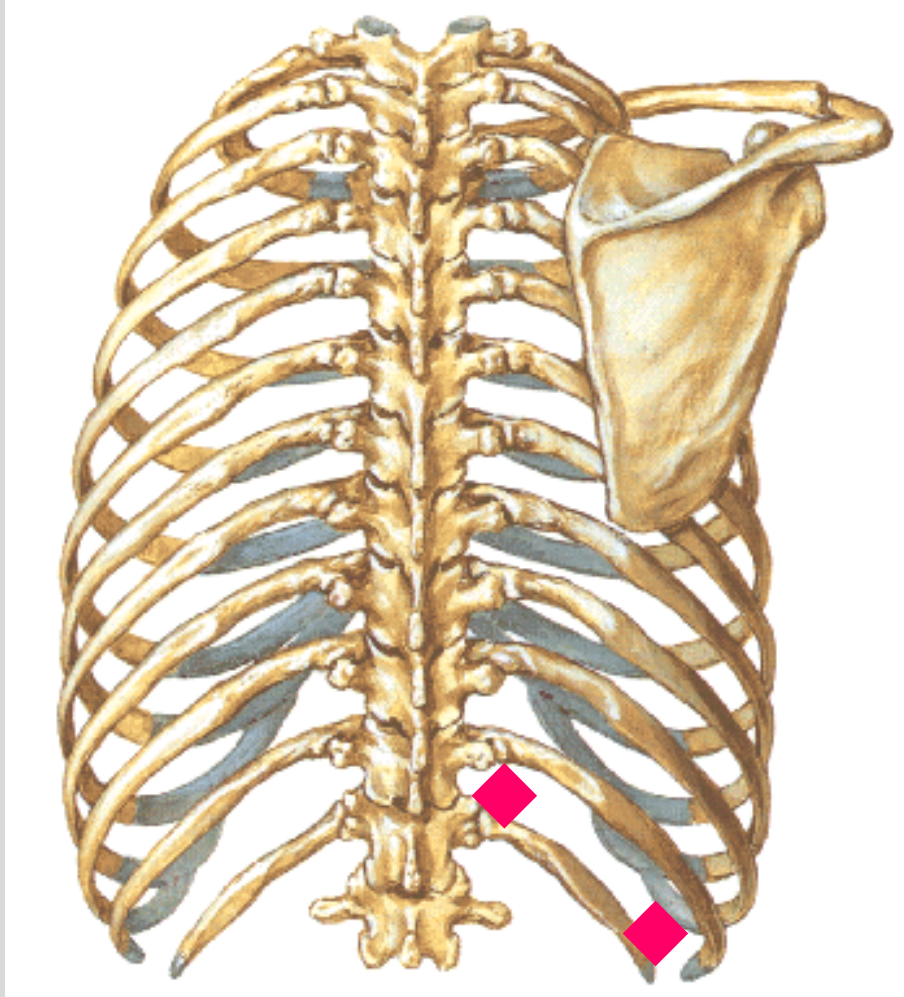
**Left Intertransverse
space between T7 & T8**



Splenic points
important for
infection



Appendix



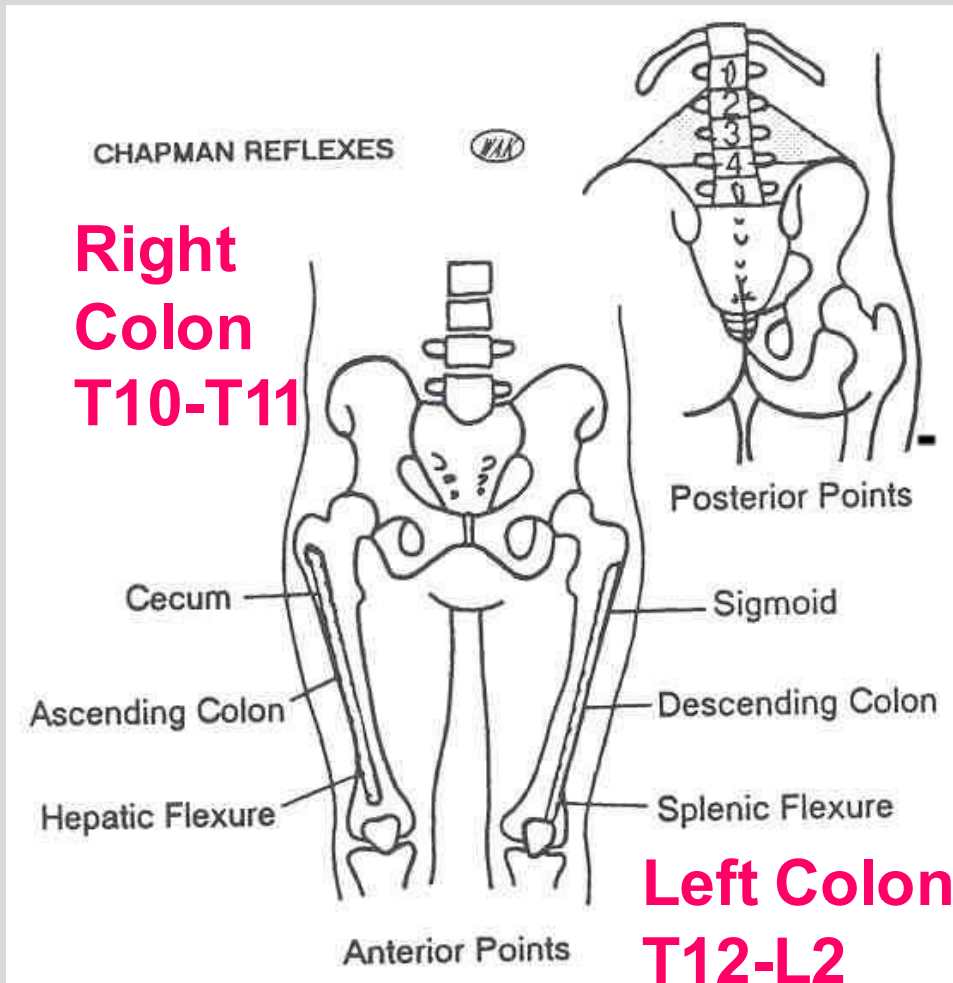
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- **Anterior point:**
Right tip of 12th rib
- **Posterior point:**
Right 11th
intertransverse space



Lower Gastrointestinal



Anterior Points

Cecum: Right Upper Outer Thigh

Ascending Colon: Right Middle Thigh

Transverse Colon 1st 2/5th (Hepatic Flexure): Right Lower Outer Thigh

Transverse Colon Last 3/5th (Splenic Flexure): Left Lower Outer Thigh

Sigmoid: Left Upper Outer Thigh

Descending Colon: Left Middle Thigh

Posterior Points

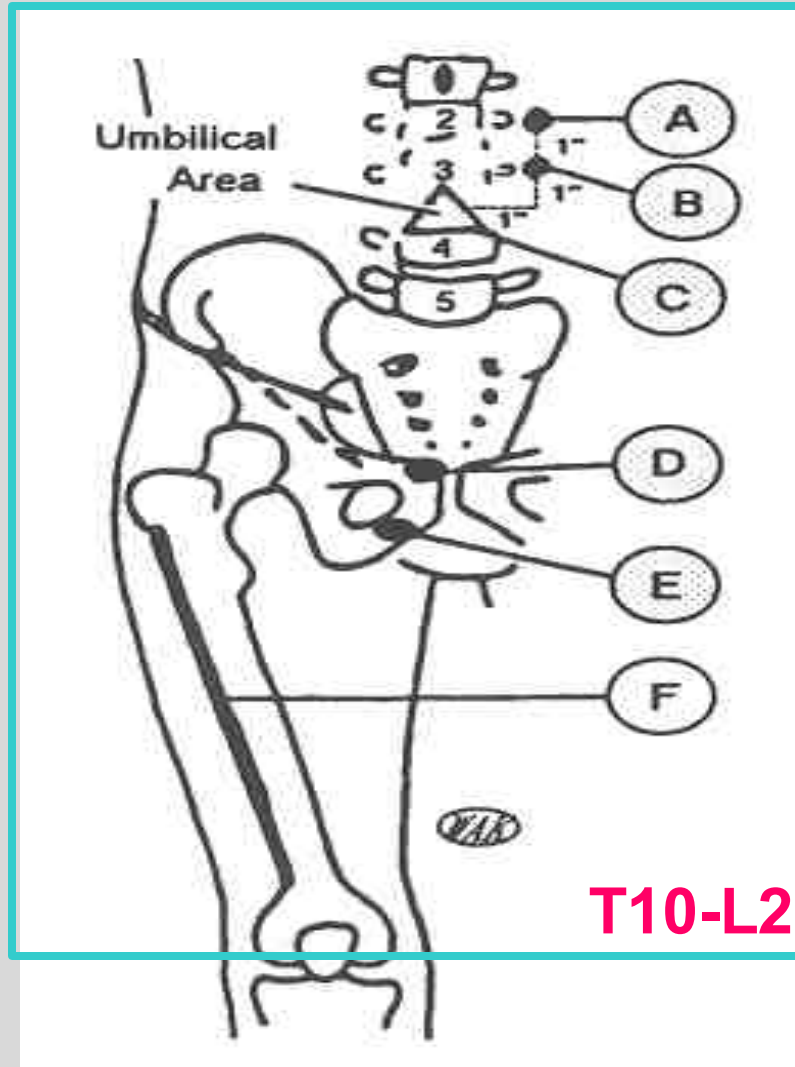
Transverse process of L2 through transverse process of L4 across crest of ilium creating a triangle.



Genitourinary

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



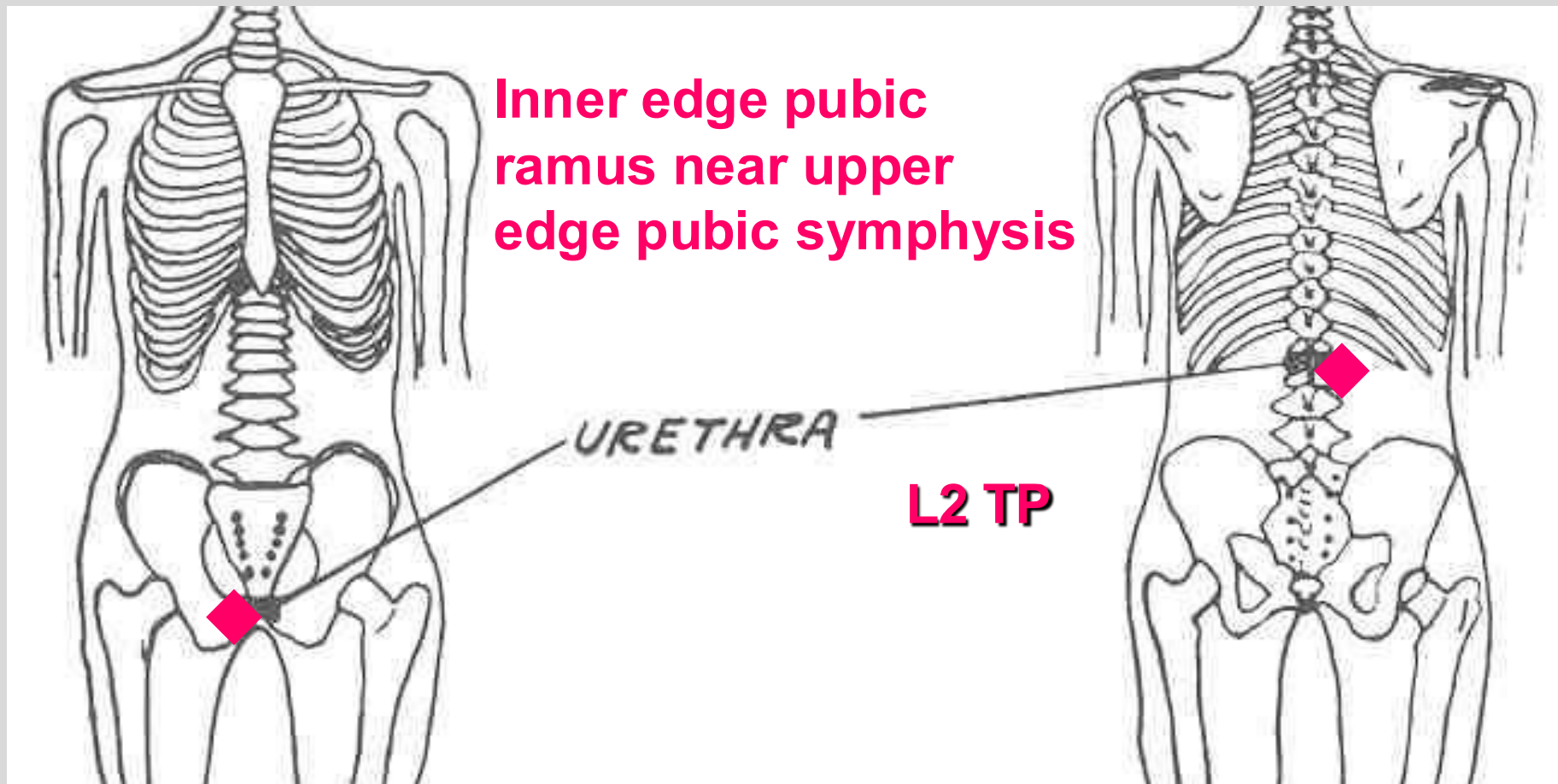
- A. Adrenals
- B. Kidney
- C. Bladder
- D. Ovaries / Gonads
- (*Urethra: between D and E)
- E. Uterus
- F. Broad Ligament / Prostate



Urethra

NEW YORK INSTITUTE
OF TECHNOLOGY

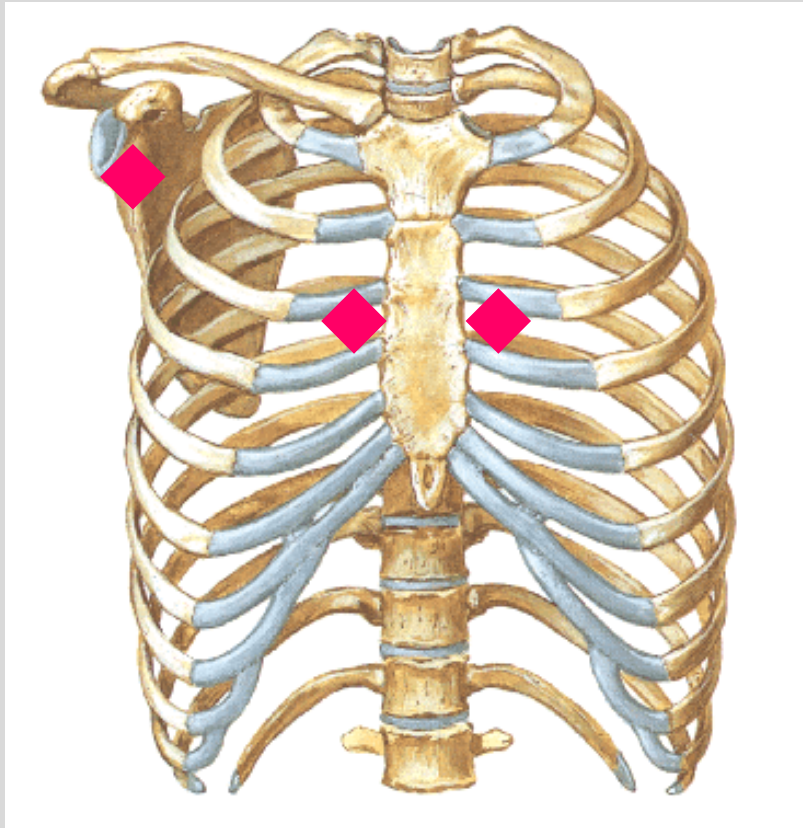
College of Osteopathic
Medicine



Neuritis of Upper Extremity

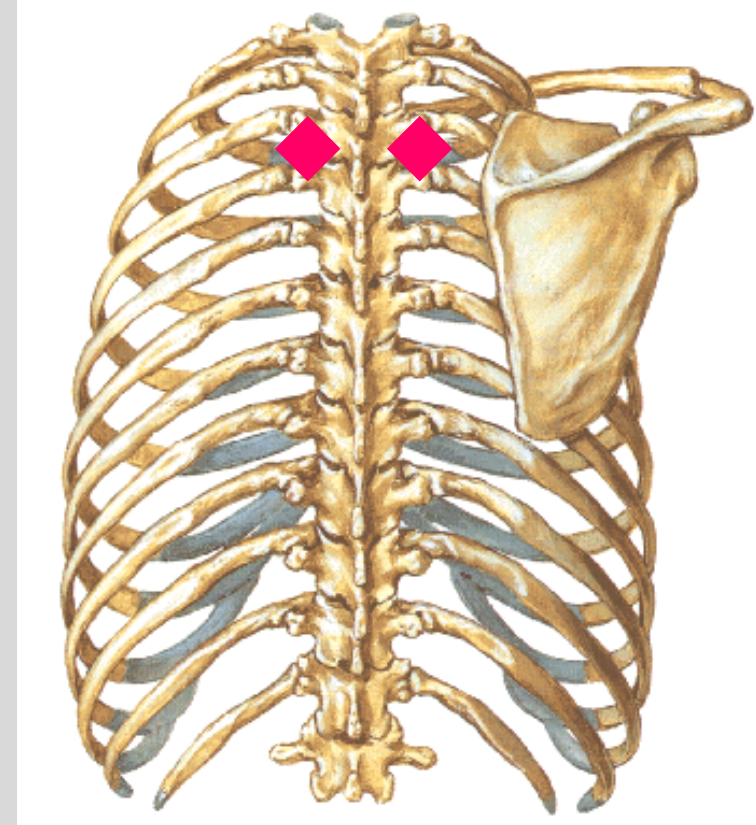
NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



**3rd Rib &
Parker's Reflex**

Department of Osteopathic Manipulative Medicine



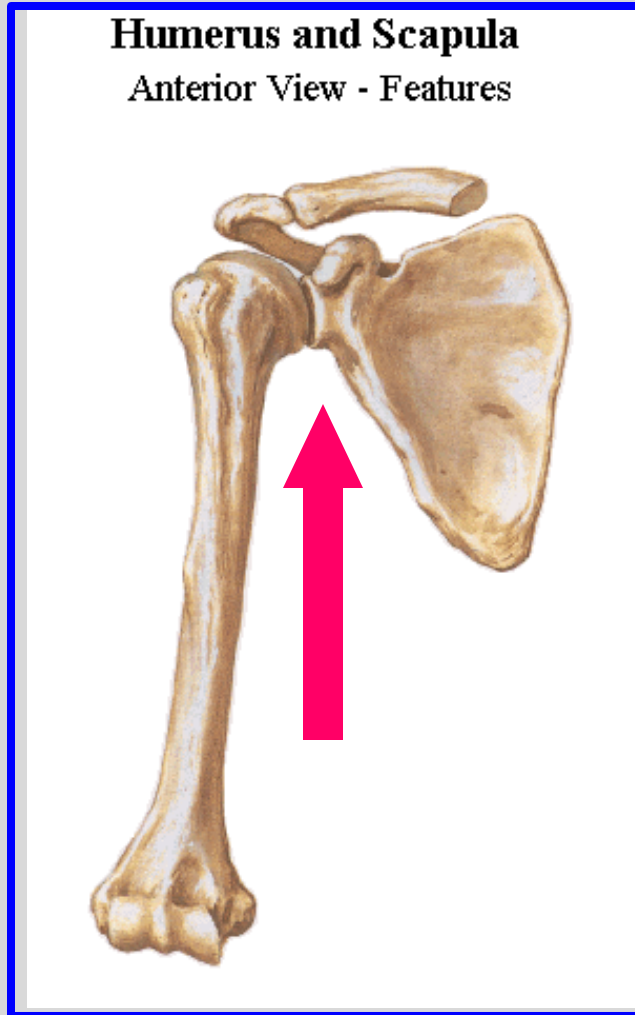
Between TP's of T3 & T4



Parker's Reflex

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



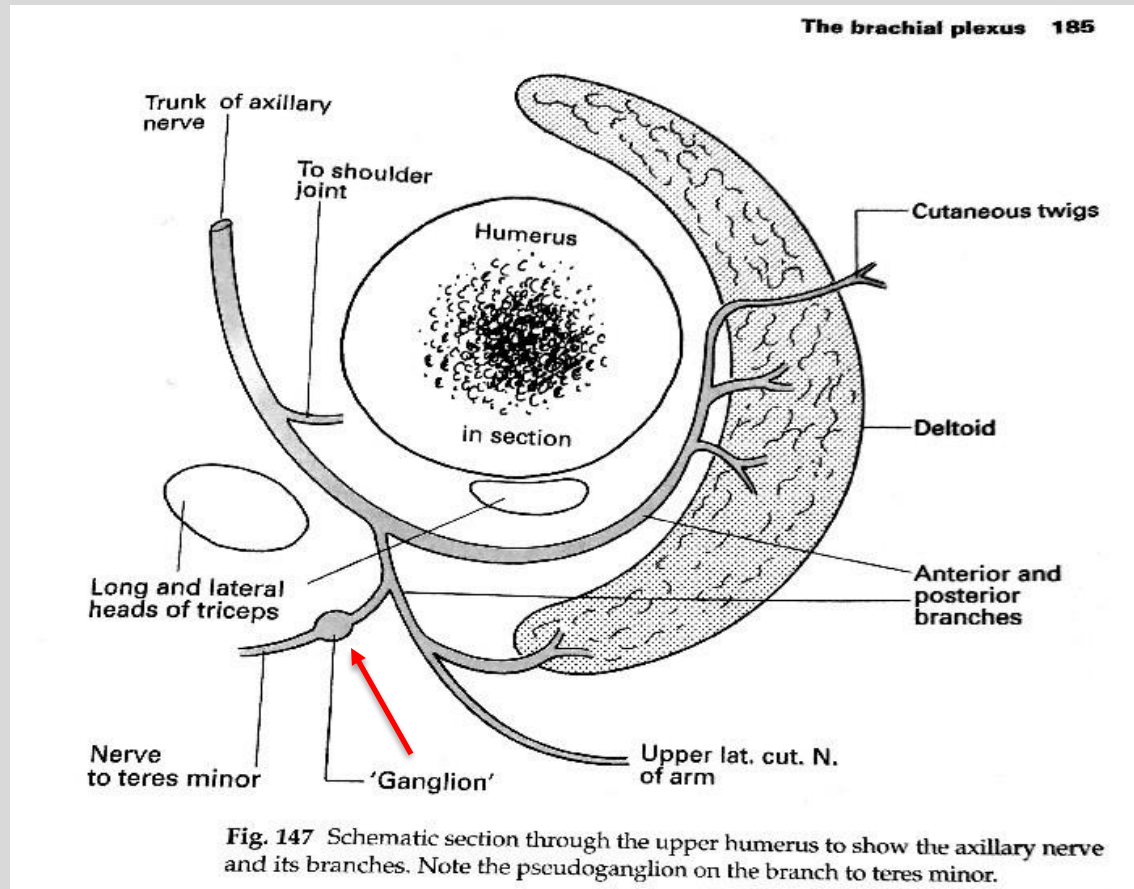
- **Found on upper outer anterior aspect of scapula. Useful in disorders from the shoulder to the hand. For example: bursitis, carpal tunnel syndrome, Dupuytren's contracture, reflex sympathetic dystrophy. Stimulates lymph flow and normalizes sympathetic tone to the upper limb.**



Note the “ganglion” in the Axilla

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Parker's reflex is a hidden gem in the Chapman's manual. Look for it on the anterior, superior and lateral border of the scapula.



PTAS: Pelvic-Thyroid-Adrenal Syndrome: The Neuro-Endocrine Immune Response: NEI

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

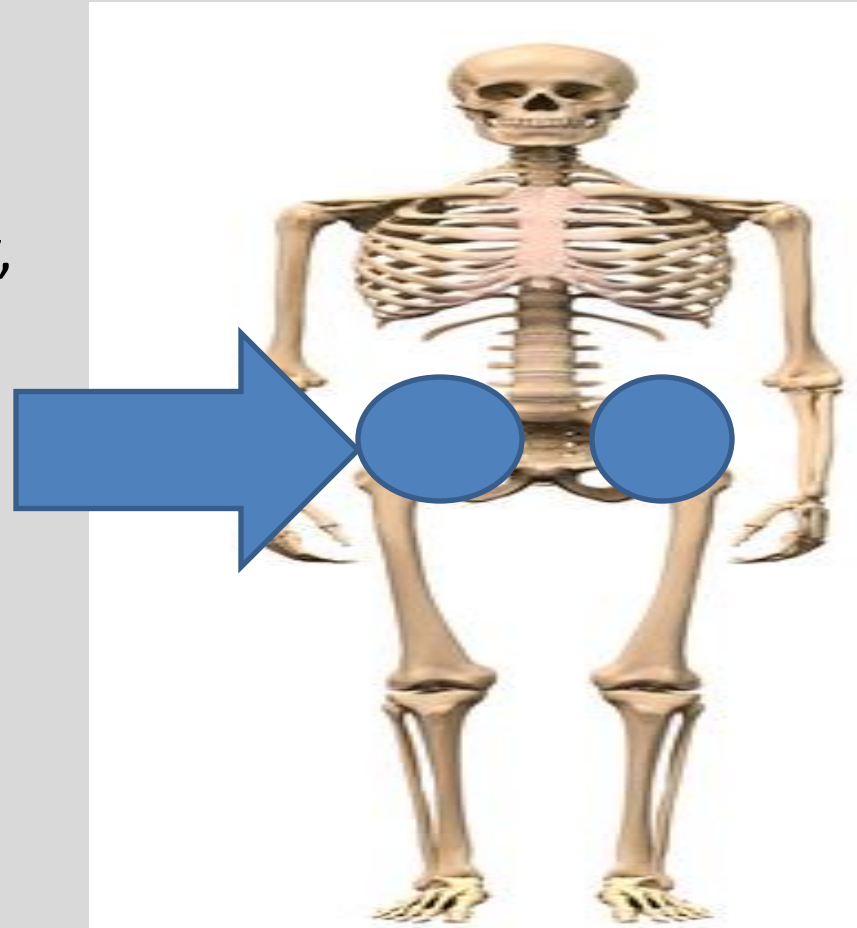
- Chapman's PTAS and its interaction with the neuroendocrine immune response is first treat the pelvis because that is where the gonads and adrenals are housed—they emit hormones such as estrogen, progesterone, testosterone, cortisol and norepinephrine for example. These organs are innervated by the sympathetics. There is a feedback cycle with thyroxine, a thyroid hormone which has widespread effects throughout the entire body, including on the immune system.
- An example of the PTAS/NEI response would be thyrotropin and thyroxine stimulates natural killer cells and produces T-cells respectively. Stress, or the musculoskeletal ready state, is mediated by the sympathetic nervous system and promotes release of cortisol from the adrenal glands, which also modulates immunocytes.

Capobianco, John D., The Neuroendocrine-immune complex as illustrated in the work of Dr. Frank Chapman. J Am Acad Osteopath. 2004; 14(1)



In All Chapman's Reflexes First Treat the Innominate: Innominate Dysfunction

Although the sacroiliac is the key dysfunction in the body, Treating the innominates are the first item of business to Do when treating Chapman's reflexes. The pelvis houses Endocrine (reproductive) glands and the terminus of the Sympathetic nervous system (ganglion impar).



Chapman's Quick Reference

HEENT

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Organ/Disease Examples	Anterior and Posterior Locations
Brain-Cerebrum/CVA/Paralysis New: Insomnia-Surface of 4 th Rib underlying scapula	A: Lateral aspect of spinous processes of C3-5 P: Between C1-2 t-processes
Brain-Cerebellum/Confusion/Childhood Learning	A: Corocoid process P: C1 t-process
Ears/Otitis media/Motion Sickness/Mastoiditis/Vertigo	A: Medial-superior clavicle P: C1 t-process
Eyes/Conjunctivitis/Retinitis	A: Surgical neck of humerus P: Occiput-suboccipital nerve
Nose/Rhinitis/Hay Fever	A: 1 st rib & cartilage P: Lateral C1 from line of jaw angle
Neurasthenia/Fatigue/Decreased Mental Alertness/Weakness Especially after Exertion. Associated with thyroid, heart, GI and Reproductive symptoms	A: Pectoralis major from humerus, clavicle, sternum, cartilage, especially down to fourth rib in midaxillary line P: 4 th Rib under scapula (associated with insomnia)
Pharynx-Eustachian tube-Uvula/Pharyngitis	A: 1 st rib $\frac{3}{4}$ inch from sternum P: Trans. Process of C2

Chapman's Quick Reference HEENT (Continued)

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Organ/Disease Examples	Anterior and Posterior Points
Tonsils/Tonsillitis-Strep throat	A: 1 st Intercostal space (ICS) P: Trans. Process C1
Tongue/Choking/Soreness	A: 2 nd rib cartilage $\frac{3}{4}$ inch from sternum P: Between spinous and transverse process of C2
Larynx/Laryngitis/Aphonia	A: 2 nd rib-superior aspect- $\frac{3}{4}$ inch from sternum P: Between spinous & transverse process of C2
Paranasal Sinuses/Sinusitis	A: 1 st ICS & 2 nd rib $\frac{3}{4}$ inch from sternum P: Between spinous & transverse process of C2
New Sinus Points	Fourth rib and lateral edge of scapula
New HEENT Congestion	2nd rib in axilla
New Acne/Psoriasis Points	3 rd and 4 th ribs posteriorly
New Epistaxis Point	Suboccipital nerve on occiput

Chapman's Quick Reference Cardiopulmonary

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Organ/Disease Examples	Anterior and Posterior Points
Heart/Myocarditis	A: 2 nd ICS P: Between Trans. processes of T2-3
Bronchials/bronchitis	A: 2 rd ICS P: Trans. processes of T2
Upper Lung/pneumonia/TB/asthma	A: 3 th ICS P: Between Trans. processes of T3-4
Lower Lung/pneumonia/TB/asthma New: Pleurisy for Lower Lung Point	A: 4 th ICS P: Between Trans. Processes of T4-5

Chapman's Quick Reference

Upper/Middle GI

**NEW YORK INSTITUTE
OF TECHNOLOGY**

Organ/Disease Examples	Anterior and Posterior Points
Esophagus/GERD/Nausea	A: 2 nd ICS P: Trans. Process T2
Stomach/Acidity/Gastric Ulcer/Neoplasm	A: 5 th ICS (L) P: Intertransverse space T5&6
Stomach/Peristalsis/Associated w/ Heart Palpitations	A: 6 th ICS (L) P: Intertransverse space T6&7
Pyloris/Pyloric Stenosis/Abdominal Bloating	A: Sternum (from junction of manubrium to junction of xyphoid) P: Trans. Process T10 (R) & 10 th Rib (R)
Pancreas/Diabetes/Glycosuria New: Pancreatic Point for Food Allergies & Bodily pH	A: 7 th ICS (R) P: Between trans. processes T7&8 (R)
Gall bladder/Gall stones/Cholecystitis New Point: Anterior/Medial Radial Head (R)	A: 6 th ICS (R) P: Between trans. processes T6&7 (R)
Liver/Enlargement	A: 6 th ICS (R) P: Between trans. processes T6&7 (R)
Liver/Torpid(Sluggish)	A: 5 th ICS (R) P: Between trans. processes T5&6 (R)
Spleen/Leukemia/Infections	A: 7 th ICS (L) P: Between trans. processes T7&8 (L)
Small Intestines/Enteritis	A: 8,9&10 th ICS P: Between trans. processes T8-9,9-10&10-11

Chapman's Quick Reference

Lower GI

Organ/Disease Examples	Anterior and Posterior Points
Appendix/Appendicitis New: 7 o'clock on right border of umbilicus (in line with McBurney's point)	A: Tip of anterior portion of 12 th rib (R) P: Trans. Process of T11 ®
Iliocecal Valve-Cecum/Colitis/Spastic Constipation	A: Greater Trochanter (R) (Upper Fifth of Femur) (Lateral Thigh) P: Trans. Process of L2,3&4 to Iliac crest (Applies to All Colon Points)
Ascending Colon/Colitis/Spastic Constipation	A: Lateral Femur (R) (Middle Femur) (Lateral Thigh) P: Same as above
Transverse Colon/Hepatic Flexure (First 2/5 th s of Transverse Colon)/Colitis/Spastic Constipation	A: Lower Femur (R) (Last Fifth of Femur) (Lateral Thigh) P: Same as above
Transverse Colon/Splenic Flexure (Last 3/5 th s of Transverse Colon) (First Lower Fifth of Femur)/Colitis/Spastic Constipation	A: Lower Femur (L) (Last Fifth of Femur) (Lateral Thigh) P: Same as above
Descending Colon /Colitis/Spastic Constipation	A: Lateral Femur (L) (Middle 3/5 th of Femur) (Lateral Thigh) P: Same as above
Sigmoid Colon/Colitis/Spastic Constipation	A: Lateral Femur (L) (Upper Femur-Greater Trochanter-L) (Upper fifth of Lateral Thigh) P: Same as above
Rectum	A: Lesser Trochanter P: Sacrum-Lower end of Iliosacral Joint)
Hemorrhoids	A: Above Ischial Tuberosities P: Sacrum-Close to ilium at Iliosacral Joint)
Atonic Constipation/Dilation	A: Between ASIS and Greater Trochanter P: Trans. Process of T11 and 11 th Rib

Chapman's Quick Reference Genitourinary

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Organ/Disease Examples	Anterior and Posterior Points
Kidneys/Bright's Ds (Edema, HTN and albuminuria)/anuria/polyuria/calculi/abscess/neoplasm	A: 1 inch lateral & 1 inch superior to umbilicus P: Between trans. process of T12&L1
Urinary Bladder/Cystitis-infection (urinary (frequency)	A: Umbilicus & Pubic Symphysis-B/T upper & lower pubes P: Upper edge of trans. process of L2
Urethra (Cystitis or calculi)	A: Inner edge of pubic ramus near upper edge of pubic symphysis P: Upper edge of trans. process of L2
New: Enuresis (Bedwetting)	Anterior Only: Pubic Symphysis

Chapman's Quick Reference Reproductive

**NEW YORK INSTITUTE
OF TECHNOLOGY**

Organ/Disease Examples	Anterior and Posterior Points
Prostate/Prostatitis New: STD causing Prostatitis	A: Lateral Femur from Gr. Trochanter to 2 inch above knee A: Lateral margin of pubic symphysis P: Between spinous process of L5 and PSIS
Orchitis//Epididymitis/Contraction of Inguinal rings Note: This is known as the Abdominal Tension point	A: Below inguinal (Poupart's) ligament upper edge of body/ramus of pubes P: Tip of trans. process of L2
Ovaries/Ovarian cyst	A: Round ligaments from upper to lower border pubic bone P: Inner half of ovary between trans. process of T9&10 P: Outer half of ovary between trans. process of T10&11
Seminal vesicles/Vesiculitis (From STD) Fallopian Tubes//Salpingitis/Pyrosalpinx (From STD)	A: Between acetabulum and sciatic notch P: Between spinous process of L5 and PSIS
Leucorrhea/Vaginal secretions (May be from infection or normal)	A: Medial femur condyle & 3-6 inches on posterior aspect (Near knee) P: Between spinous process of L5 and PSIS
Clitoris/Priapism Vagina/Vaginismus	A: Upper/Inner/Posterior thigh 3-5 inch vertical to 1 ½-2 inches lateral P: Sacrococcygeal joint
Uterus/Inflammation	A: Upper edge junction of ramus of pubes with ischium P: Between spinous process of L5 and PSIS
Uterus/Fibroma/Neoplasm New: Breast Angle of ribs 5-6 (May be associated with uterine neoplasm)	A: Pubic symphysis to obturator foramen P: Tip of trans. process of L5 to iliac crest
Broad Ligament (of Uterus/Ovaries/Fallopian Tubes)/Abnormal Position of Uterus New: STD causing inflammation of Broad Ligament	A: Lateral femur from Gr. Trochanter to 2 inches above knee joint P: Between spinous process of L5 and PSIS
Groin Glands/Lymph nodes of femoral/inguinal area draining pelvis)/Infection	A: Inner condyle of knee lower 2/5ths of sartorius muscle to tibia P: Iliosacral joint

Chapman's Quick Reference Endocrine

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Organ/Disease Examples	Anterior and Post
Thyroid/Thyroiditis/Hyper/Hypo	A: 2 nd ICS P: Trans. process of T2
Adrenals/Fatigue	A: 1 inch lateral and 2-2 ½ inches superior to umbilicus P: Between trans. Process of T11&12
Innominate-Always treat innominate as dysfunction signify endocrinopathy Always treat innominate before any Chapman's points	A: Thickened, sensitive inguinal (Poupart's) ligament associated with a posterior innominate and or leg length discrepancy

Chapman's Quick Reference Musculoskeletal

**NEW YORK INSTITUTE
OF TECHNOLOGY**

Organ/Disease Examples	Anterior and Posterior Points
Arms/Poor Circulation/Numbness/Tingling	A: Pectoralis minor to attachments of Ribs 3-5 P: Superior angle of scapula and ribs 1-3, especially inner margin of scapula
Upper Limb/Neuritis	A: 3 rd ICS P: Between trans. process of T3&4 and a depressed 3 rd rib (A depression of the third rib is also seen with disorders of the foot)
Upper Extremity/Hand//Dupuytren's Contracture-PARKER'S REFLEX	A: Upper, anterior, superior edge of scapula below head of humerus
Wry Neck/Torticollis/Cervical Dystonia	A: Ant. Humerus surgical neck&caudad P: Trans. Processes C3,4,6&7 New points include C2-4, T1-4 & ribs 1-2
Sciatica/Sciatic Neuritis/Lumbosacral Radiculopathy/Impingement of sciatic nerve along piriformis	A: Posterior/Outer/Lateral Femur 2-3 inches below Gr. Trochanter A: 1/5 distance above knee for 2 inches on post. Lateral femur A: 1/3 upward from condyles to mid-posterior lateral femur A: Supplemental: Superior fibula down to lateral malleolus A: Supplemental: Midway b/t trochanter and ischial tuberosities and above trochanters A: Supplemental: Below PSIS P: Upper aspect of Sacroiliac joint Note: Innominate dysfunction will be found

References:

1. Arbuckle BE. Reflexes, The Selected Writings of Beryl E. Arbuckle, D.O., F.A.C.O.P. Rev. ed. Indianapolis, IN: American Academy of Osteopathy; 1994.
2. Beal MC. Viscerosomatic reflexes: a review. *J Am Osteo Assoc*. 1985;85(12):786–801.
3. Brown EA. Clinical Aspects of the Chapman's Reflexes. In: Northup TL, ed. Academy of Applied Osteopathy Yearbook. Ann Arbor, MI: Edwards Brothers, Inc; 1949.
4. Capobianco, J.D. (2021) Chapman's Reflex Points. In: DiGiovanna, E.L., Amen, C. and Burns, D.K., An Osteopathic Approach to Diagnosis and Treatment. 4th Ed.. Lippincott Williams & Wilkins
5. Kuchera ML, Kuchera WA. Osteopathic Considerations in Systemic Dysfunction. Rev. 2nd ed. Columbus, OH: Greyden Press; 1994.
6. Mannino JR. The application of neurologic reflexes to the treatment of hypertension. *J Am Osteopath Assoc*. 1979 Dec;79(4):225-31. PMID: 583146.
7. Owens C. An Endocrine Interpretation of Chapman's Reflexes (1937). Carmel, CA: Academy of Applied Osteopathy; 1969 (reprint).
8. Rivera-Martinez, S, Capobianco, JD. Osteopathic Treatment of Nephrotic Syndrome. *The American Academy of Osteopathy Journal*, Volume 11, Number 3, Fall, 2001; 24-28
9. Thorpe RG. Osteopathic Manipulative Therapy for Infections. *Osteopathic Annals*. New York, NY: Insight Publishing Co., Inc; 1980.
10. Thorpe RG. Psychodynamics of Stress and Relationships with the Musculoskeletal System. *Osteopathic Annals*. New York, NY: Insight Publishing Co., Inc; 1973.
11. Seffinger, D. M. (2018). *Foundations of osteopathic medicine* (4th ed.). Lippincott Williams and Wilkins., pp.224, 966-980
12. Washington, K, Mosiello, R, Venditto, M, Simelaro, J, Coughlin, P, Crow, WT and Nicholas, A. "Presence of Chapman reflex points in hospitalized patients with pneumonia" *Journal of Osteopathic Medicine*, vol. 103, no. 10, 2003, pp. 479-483. <https://doi.org/10.7556/jaoa.2003.103.10.479>
13. Willard FH, Mokler DJ, Morgane PJ. Neuroendocrine-immune system and homeostasis. In: Ward RC, ed. *Foundations for Osteopathic Medicine*. Baltimore, MD: Lippincott Williams and Wilkins; 1997.

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine